General Relativity

Leonard Susskind

Core track, 2008 Fall Einsteins General Theory Of Relativity
Lecture notes organized by LazyingArt LLC with Video2Book

Original lectures by Leonard Susskind. Lecture notes organized by LazyingArt LLC with Video2Book.

Contents

1 Gravity, Free Fall, and Newton's Theorem **1**

1.1 Gravity as a Special Force and Why We Start With Newton . . . 1

1.2 Galileo's Flat-Earth Gravity and the First Equivalence-Principle Lesson . . . 2

1.2.1 Question & Answer . . . 3

1.3 Newton's Inverse-Square Force: From Magnitude to Direction . . . 4

1.3.1 Question & Answer . . . 5

1.4 From Two Bodies to Many Bodies . . . 6

1.5 Tidal Forces as the Repair to Uniform Free Fall . . . 7

1.5.1 Question & Answer . . . 7

1.6 The Gravitational Field . . . 8

1.7 Divergence, Gauss's Theorem, and Newton's Theorem . . . 8

1.7.1 Question & Answer . . . 11

1.8 Summary . . . 12

2 Newtonian Gravity, Gauss's Law, and the Equivalence Principle **13**

2.1 Dark Energy as a Prelude, Not the Main Road . . . 13

2.1.1 Question & Answer . . . 14

2.2 A Little Formalism: Del, Fields, and the Gravitational Acceleration . . . 14

2.3 Gauss's Law as Physics, Gauss's Theorem as Mathematics . . . 15

2.4 Putting Them Together: Exterior Fields and the Shell Theorem . . . 17

2.4.1 Question & Answer . . . 17

2.5 While We Are At It: Inside the Earth . . . 18

2.5.1 Question & Answer . . . 19

2.6 The Equivalence Principle as Elevator Intuition and Coordinate Transformation . . . 20

2.7 Einstein's Move: Light, Locality, and the Metric . . . 21

2.7.1 Question & Answer . . . 23

2.8 Summary 24

3 Vertical Light Rays, Tensors, and the Metric 25

3.1 Vertical Light Rays, Doppler Shift, and the Equivalence Principle 25

3.1.1 Question & Answer 26

3.2 Free Fall, Local Cancellation, and Tidal Obstruction 26

3.2.1 Question & Answer 27

3.3 The Minimal Calculus Toolkit for Curved Coordinates 27

3.4 Scalars, Vectors, and Coordinate Dependence of Components 28

3.5 From Vectors to Tensors, and from Contravariant to Covariant 29

3.6 The Metric Tensor from Pythagoras 30

3.7 Flatness, Curvature, and the Meaning of an Obstruction 32

3.8 Polar Coordinates as a Worked Flat Example, Then the Sphere 34

3.8.1 Question & Answer 35

3.9 Summary 35

4 Tensor Algebra, Metrics, and the First Step to Space-Time 36

4.1 Covariant and Contravariant Review 36

4.1.1 Question & Answer 37

4.2 From Vectors to General Tensors 37

4.2.1 Question & Answer 38

4.3 The Metric Tensor as Distance 38

4.3.1 Question & Answer 39

4.4 Metric as Matrix, Inverse Metric, and Raising/Lowering 39

4.4.1 Question & Answer 41

4.4.2 Question & Answer 42

4.5 Tensor Equalities and the End of Tensor Algebra 43

4.6 Flat Space, Cartesian Frames, and the Pivot to Space-Time 44

4.7 General Space-Time Metric and the First Cosmological Example 46

4.7.1 Question & Answer 47

4.8 Summary 47

5 Tensor Calculus, Connections, and Geodesics 48

5.1 Why Tensor Calculus Enters General Relativity 48

5.2 Scalars First, Then the Trouble with Vectors 49

5.3 Testing the Ordinary Derivative of a Vector 49
5.3.1 Question & Answer 50
5.4 Covariant Derivatives of Vectors and Tensors 51
5.4.1 Question & Answer 52
5.5 A Concrete Example: Polar Coordinates 53
5.6 Metric Compatibility and the Christoffel Symbols 53
5.7 Differentiation Along a Curve 54
5.8 Geodesics and the Road Toward Gravity 55
5.9 Summary 56

6 Proper Time, Local Flatness, and Curvature from Parallel Transport 57
6.1 Space-Time Metric, Proper Time, and the Minkowski Signature 57
6.1.1 Question & Answer 58
6.2 Curved Coordinates Are Not Yet Curvature 59
6.3 What Does It Mean for Space or Space-Time to Be Flat? 60
6.4 Local Coordinate Freedom and Vanishing First Derivatives 60
6.4.1 Question & Answer 62
6.5 Christoffel Symbols and Metric Compatibility 62
6.5.1 Question & Answer 63
6.6 Covariant Derivative Along a Curve, Geodesics, and Parallel Transport 63
6.6.1 Question & Answer 65
6.7 Curvature as Path Dependence of Parallel Transport 65
6.7.1 Question & Answer 67
6.8 Summary 68

Chapter 1

Gravity, Free Fall, and Newton's Theorem

Gravity enters this course already carrying a double burden. It is one force among others, but it is also the force that seems to know something about geometry. Susskind therefore begins with a deliberate narrowing of scope. We are not yet doing the full general theory of relativity. We are doing Newtonian gravity, but we are doing it in a way that will later make the passage to general relativity feel motivated rather than abrupt: first the Newtonian language of forces and vectors, then the cancellation of mass in gravitational motion, then the full inverse-square law, and finally the vector calculus that explains why spherical gravitating bodies behave as if all their mass were concentrated at a point.

Source note. These companion notes follow Leonard Susskind's first general relativity lecture and preserve its order of argument. This draft was prepared with support from LazyingArt LLC and Video2Book.

1.1 Gravity as a Special Force and Why We Start With Newton

Susskind opens by saying, in effect, that gravity is unusual. Electric and magnetic forces do not behave like it, and somehow gravity is tied to the geometry of space and time. But precisely because of that, he does not begin with curved spacetime. He begins with Newton.

Definition 1.1. An inertial frame is a coordinate system, equipped with clocks, in which a force-free test object moves with uniform velocity and therefore has zero acceleration.

This is the lecture's first piece of discipline. Before we speak about gravity, we must know what we mean by force, acceleration, coordinates, and vectors. The preserved frame below catches the board at the moment when Susskind has written only the bare scalar-looking equation $F = ma$. The transcript makes clear, however, that he is already speaking of a three-dimensional vector equation.

Thus we write

$$\vec{F} = m\vec{a}, \tag{1.1}$$

$$F_i = ma_i = m\ddot{x}_i, \qquad i = 1, 2, 3. \tag{1.2}$$

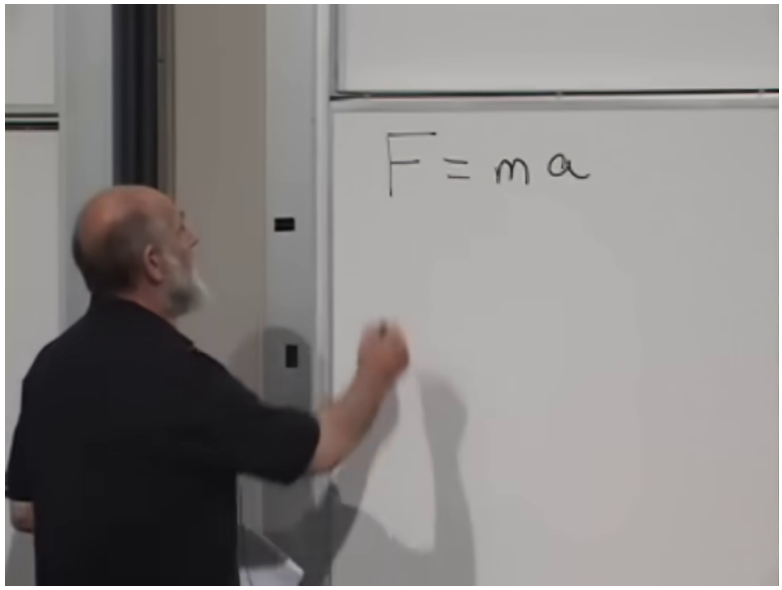


Figure 1.1: The lecture begins from the stripped-down Newtonian law. The board still shows the scalar handwriting $F = ma$, but the spoken lecture already treats it as the vector equation $\vec{F} = m\vec{a}$.

The mass m is a scalar. In Newtonian physics it is conserved: it may be redistributed from one place to another, but it is not itself a directional object.

A particle has coordinates x_1, x_2, x_3, which together form the position vector

$$\vec{r} = (x_1, x_2, x_3). \tag{1.3}$$

Its first time derivative is velocity and its second time derivative is acceleration:

$$v_i = \dot{x}_i, \qquad a_i = \ddot{x}_i. \tag{1.4}$$

This is the quiet mathematical framework within which the special feature of gravity will soon appear. If the gravitational force carries the same mass that already sits in Newton's law, then that mass may cancel, and when it cancels the motion stops depending on the nature of the falling object.

1.2 Galileo's Flat-Earth Gravity and the First Equivalence-Principle Lesson

Now the lecture deliberately steps backward from Newton to Galileo. This is not antiquarianism. Galileo isolates the simplest approximation in which gravity reveals what is peculiar about it.

In the flat-Earth approximation two simplifications hold at once. First, the direction of gravity is everywhere the same: downward. Second, its magnitude does not depend on height. If we choose the x_2-axis to be vertical and take upward to be positive, then the gravitational force has only one nonzero component:

$$F_2 = -mg. \tag{1.5}$$

The minus sign merely says that the force points downward. Here g is the near-surface gravitational acceleration. On Earth it is about $10\,\mathrm{m/s^2}$; on the Moon it is smaller, on Jupiter larger. What

matters in the present argument is that for a given planet it does not depend on the mass of the falling body.

Combining Galileo's force law with Newton's equation gives the first decisive cancellation:

$$F_2 = m\ddot{x}_2, \tag{1.6}$$

$$m\ddot{x}_2 = -mg, \tag{1.7}$$

$$\ddot{x}_2 = -g. \tag{1.8}$$

This is the first serious mathematical signal of the equivalence principle in the lecture. The acceleration of free fall is independent of the mass of the object. Heavy bodies and light bodies obey the same equation of motion. However Galileo performed the experiment, whether from a tower or on an inclined plane, this is the content of the result: absent air resistance, different masses fall alike.

The crucial point is that this is not true for generic forces. Electric forces couple to charge, not mass. Gravity is special because its force is proportional to the same mass that Newton's law then multiplies by the acceleration. The mass cancels, and with it goes any dependence of the motion on the identity of the falling body.

1.2.1 Question & Answer

Why can a freely falling cloud fail to detect gravity even though gravity is present?

Take a diffuse cloud of particles and let them all go in the same uniform gravitational field. Some particles may be heavy, some light, some large, some small. None of that matters. Every particle obeys $\ddot{x}_2 = -g$. Therefore every particle falls in exactly the same way.

The consequence is stronger than the slogan "all bodies fall alike." The cloud does not deform. Neighboring particles keep the same relative separation. If the cloud had been held together by struts, those struts would feel no stress from the gravitational field itself, because all parts of the structure share the same motion. In the flat-Earth approximation, gravity produces no internal stretching, no compression, and no shear.

That is why Susskind says that free fall is locally indistinguishable from being far away from all gravitating bodies. The point is not that one cannot tell the difference between standing on the Earth and falling. Of course one can. The point is subtler: within a self-contained falling system, and without using an external reference such as the floor, one cannot detect the gravitational field by internal mechanical measurements.

The student questions in the lecture push this further. Is the effect optically detectable? Suppose the particles in the cloud carry lasers. The answer is still no, because the relative distances do not change. Is the peculiar feeling of stepping off a building evidence of gravitational acceleration? Again no. That feeling comes from the disappearance of the support forces that normally act on the bottoms of our feet. In Susskind's phrase, what we ordinarily feel is not falling but *not* falling.

This is the lecture's first version of the equivalence principle: in a perfectly uniform gravitational field, free fall erases gravity locally.

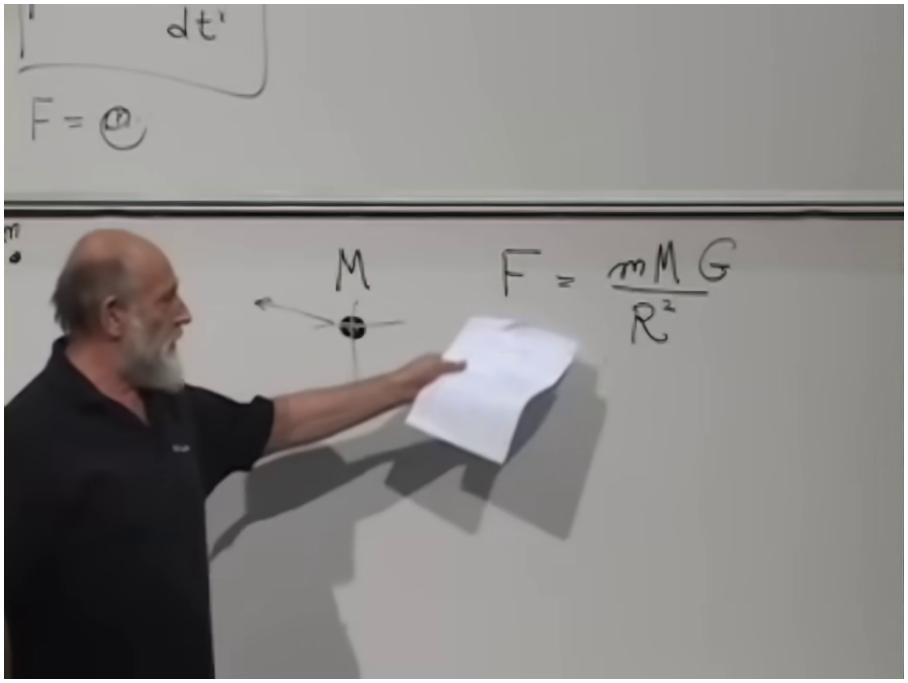


Figure 1.2: The lecture first writes Newton's law as a statement about force magnitude. The point-source sketch labeled M prepares the immediate transition to vector form.

1.3 Newton's Inverse-Square Force: From Magnitude to Direction

The flat-Earth approximation has now done its work. The lecture turns to Newton's full force law. Every object in the universe exerts a gravitational force on every other object. Start with two masses, a lighter mass m and a heavier mass M. The forces are attractive, equal, and opposite.

The board first presents the law as a statement about magnitude, not yet about direction. In the preserved frame the denominator is written as R^2; we keep that equation visible because it is legible on the board, but in the polished formulas below we normalize the notation to $r = |\vec{r}|$.

On the board one sees

$$F = \frac{mMG}{R^2}. \tag{1.9}$$

In cleaned notation we write

$$F = \frac{GmM}{r^2}, \tag{1.10}$$

where G is Newton's constant,

$$G \approx 6.7 \times 10^{-11}\,\mathrm{N\,m^2/kg^2}. \tag{1.11}$$

The lecture pauses over the smallness of this number. Gravity is weak. That is not merely rhetoric. For two 1 kg masses separated by 1 m,

$$F = G \approx 6.7 \times 10^{-11}\,\mathrm{N}. \tag{1.12}$$

A bit of electrostatic charge or a modest magnet easily beats that. Gravity dominates ordinary life only because the Earth is so heavy.

There is another special feature here as well. Electric forces may attract or repel. Gravitational forces, in Newton's theory, are always attractive.

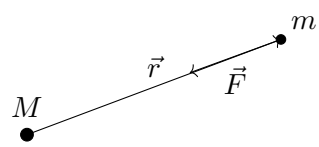


Figure 1.3: Clean reconstruction of the geometry behind the vector law. The radial vector points from the source mass M to the test mass m, while the force points inward, opposite to $\vec{r}$.

Now Susskind sharpens the scalar law into a vector law. Put the heavy mass M at the origin and let $\vec{r}$ point from M to the lighter particle m. Then the force on m must point along the radial direction, but opposite to $\vec{r}$. Introducing the unit vector

$$\hat{r} = \frac{\vec{r}}{r}, \tag{1.13}$$

we obtain

$$\vec{F} = -\frac{GmM}{r^2}\hat{r} = -\frac{GmM}{r^3}\vec{r}. \tag{1.14}$$

The extra factor of r downstairs is exactly what keeps the magnitude at $1/r^2$ while inserting the direction.

1.3.1 Question & Answer

Equal forces, unequal accelerations. If the force on the Earth due to a falling body is equal and opposite to the force on the body due to the Earth, why does the Earth hardly accelerate at all?

Because equality of forces is not equality of accelerations. Newton's equation says

$$\vec{a} = \frac{\vec{F}}{m}.$$

For the same force, the larger mass has the smaller acceleration. The Earth and the falling body exert equal forces on one another, but the Earth's enormous mass suppresses its acceleration.

Why inverse square? The lecture's historical motivation runs through Kepler. For circular motion of radius r and angular frequency ω, the inward acceleration is

$$a_{\text{circ}} = \omega^2 r. \tag{1.15}$$

If the force per unit mass behaves as $f(r) \propto 1/r^2$, then

$$\omega^2 r \propto \frac{1}{r^2}, \qquad \text{so} \qquad \omega^2 \propto \frac{1}{r^3}. \tag{1.16}$$

Since $T = 2\pi/\omega$, we obtain

$$T^2 \propto r^3. \tag{1.17}$$

This is Kepler's law relating period and orbital radius. In the lecture this is not meant as a full historical reconstruction, but as the right Newtonian clue: Kepler's law points strongly toward the inverse-square force, and the inverse-square force is then the law that extends correctly beyond perfect circles to the ellipse.

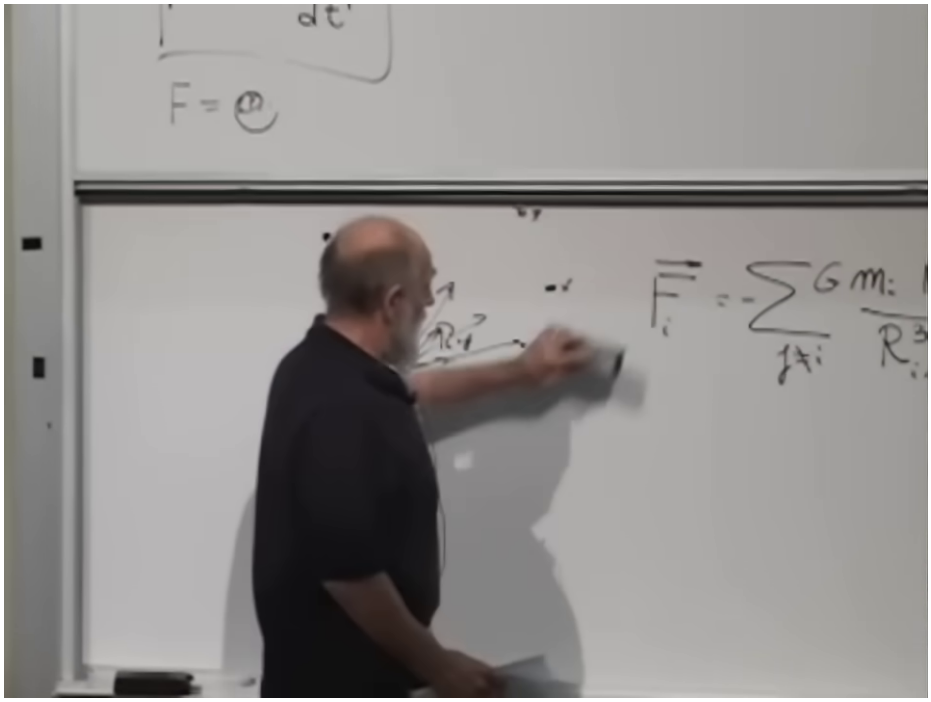


Figure 1.4: Documentary witness for the many-body discussion. The board is already being erased and the sign convention is being corrected live, so the final notes immediately replace the unstable blackboard notation by a clean standard form.

What if the bodies get too close? The lecture also inserts a caution. When bodies touch, or even come sufficiently close, gravity is no longer the whole story. Electrostatic, atomic, and nuclear forces enter. Newton's inverse-square law is the right long-range gravitational law, not a complete short-distance theory of matter.

1.4 From Two Bodies to Many Bodies

Now we pass from one pair of gravitating bodies to a whole assembly of particles. Newtonian gravity works pairwise, but the total force on a given particle is the vector sum of all the pairwise forces exerted by the others. Susskind pauses to remark that this superposition rule is not logically forced by the idea of force itself; it is a fact about Newtonian gravity.

The available board witness for this part of the lecture is weak but useful. It catches the many-body formula in the middle of a live correction, just as the sign convention is being discussed and the board is already being erased.

Let $\vec{r}_i$ be the position of the i-th particle. If we define the vector from particle i to particle j by $\vec{r}_j - \vec{r}_i$, then the clean Newtonian force law is

$$\vec{F}_i = \sum_{j \neq i} G\, m_i m_j \, \frac{\vec{r}_j - \vec{r}_i}{|\vec{r}_j - \vec{r}_i|^3}. \tag{1.18}$$

This is the simplest way to encode attraction without importing the lecture's temporary sign ambiguity. The student objection in the lecture is exactly that one may put the minus sign in

front or absorb it into the choice of whether the vector points from i to j or from j to i. Once the convention is fixed, the formula is fixed.

Now combine this with Newton's equation for the same particle:

$$\vec{F}_i = m_i \vec{a}_i, \tag{1.19}$$

$$m_i \vec{a}_i = \sum_{j \neq i} G\, m_i m_j \, \frac{\vec{r}_j - \vec{r}_i}{|\vec{r}_j - \vec{r}_i|^3}. \tag{1.20}$$

The mass m_i cancels:

$$\vec{a}_i = \sum_{j \neq i} G\, m_j \, \frac{\vec{r}_j - \vec{r}_i}{|\vec{r}_j - \vec{r}_i|^3}. \tag{1.21}$$

Once again the motion of the particle we are tracking does not depend on its own mass. It depends on the masses of all the other particles, but not on its own. This is the same structural fact we already saw in the flat-Earth problem, now stated for the general Newtonian field.

The lecture also inserts an important warning here. This formula tells us the instantaneous force if we know the positions of all the particles. Once the entire system is allowed to evolve, every particle changes the motion of every other particle, and the full dynamics quickly becomes a complicated many-body problem. That complication is real, but it is not the point of the present lecture. The present point is the persistence of mass-independence in the motion of a test object.

1.5 Tidal Forces as the Repair to Uniform Free Fall

At this point the lecture returns to the earlier statement about free fall and tightens it. In the flat-Earth approximation we said that a freely falling object does not detect gravity internally. That statement was deliberately a little too strong. It is true only for a perfectly uniform field.

Once the force depends on distance from the source, not all parts of an extended body accelerate in exactly the same way. If a very tall body falls feet-first toward the Earth, the feet are closer to the Earth than the head, so the lower part experiences a stronger acceleration. The body is stretched vertically. In differential language, nearby parts separated by $\delta\vec{x}$ experience different accelerations because

$$\delta\vec{a} \approx \vec{g}(\vec{x} + \delta\vec{x}) - \vec{g}(\vec{x}) \tag{1.22}$$

is no longer negligible.

There is a second effect. If the same extended body lies horizontally, then its left and right ends are pulled toward the Earth's center along slightly different directions. The field is radial, so those forces are not parallel. The body is therefore squeezed horizontally even if the magnitudes of the forces are nearly the same. In the lecture this becomes the vivid picture of the absurdly tall man who is stretched in one orientation and compressed in another.

These are tidal forces. They are real. They are not an artifact of coordinates. They arise because the Newtonian gravitational field varies in both magnitude and direction from point to point.

1.5.1 Question & Answer

If free fall hides gravity, why do sufficiently large falling bodies get stretched and squeezed?

Because free fall hides only the common part of the acceleration. If the field were perfectly uniform, every part of the body would share the same motion and nothing internal would reveal gravity. But a real gravitational field is not uniform. Different parts of the body sample different values and different directions of the field, and those differences are exactly what produce stress and strain.

So the lecture keeps two statements at once. A sufficiently small freely falling object does not notice gravity, because the gradient of the field across its size is negligible. A sufficiently large freely falling object does notice gravity, because that gradient is not negligible. Then one part is pulled a little harder than another, or one part is pulled in a slightly different direction than another, and the body is stretched or squeezed.

This is also why the forces are called *tidal*. The Moon exerts such forces on the Earth. The solid Earth is comparatively rigid, but the oceans are not. The water layer is therefore slightly stretched along the Earth–Moon line and squeezed in the transverse directions, producing the familiar tidal bulges.

1.6 The Gravitational Field

Only after this correction does Susskind define the gravitational field. The definition is operational. We imagine adding one more particle, a very light particle, light enough that it does not appreciably influence the motion of the others. This extra particle is the test particle.

Definition 1.2. The gravitational field at a point $\vec{x}$ is the acceleration of a sufficiently light test particle placed at $\vec{x}$.

This definition is tailored to the special property of gravity. Since the force is proportional to the mass of the test particle, the ratio of force to mass is independent of which test particle we use. We may therefore think directly in terms of acceleration rather than force. The field is the vector acceleration assigned to each point in space.

If the source masses are at positions $\vec{r}_j$, a clean Newtonian expression is

$$\vec{g}(\vec{x}) = \sum_j G\, m_j \, \frac{\vec{r}_j - \vec{x}}{|\vec{r}_j - \vec{x}|^3}. \tag{1.23}$$

The lecture temporarily denotes this acceleration by $\vec{A}$. We keep $\vec{g}$ for the gravitational field itself and reserve $\vec{A}$ for the generic vector field used in the vector-calculus discussion after the break.

The field is a vector field. It has magnitude and direction, it depends on position, and if the source masses move it depends on time as well. That is the right place for the lecture to pause. Once the field has been laid out over space, we need the mathematics that describes how such a field varies from point to point.

1.7 Divergence, Gauss's Theorem, and Newton's Theorem

After the break Susskind says openly that we now need a little fancy mathematics. The first object is a generic vector field $\vec{A}(\vec{x})$, not yet necessarily the gravitational field. We want a scalar quantity that measures whether the field is locally spreading out from a point or converging into it.

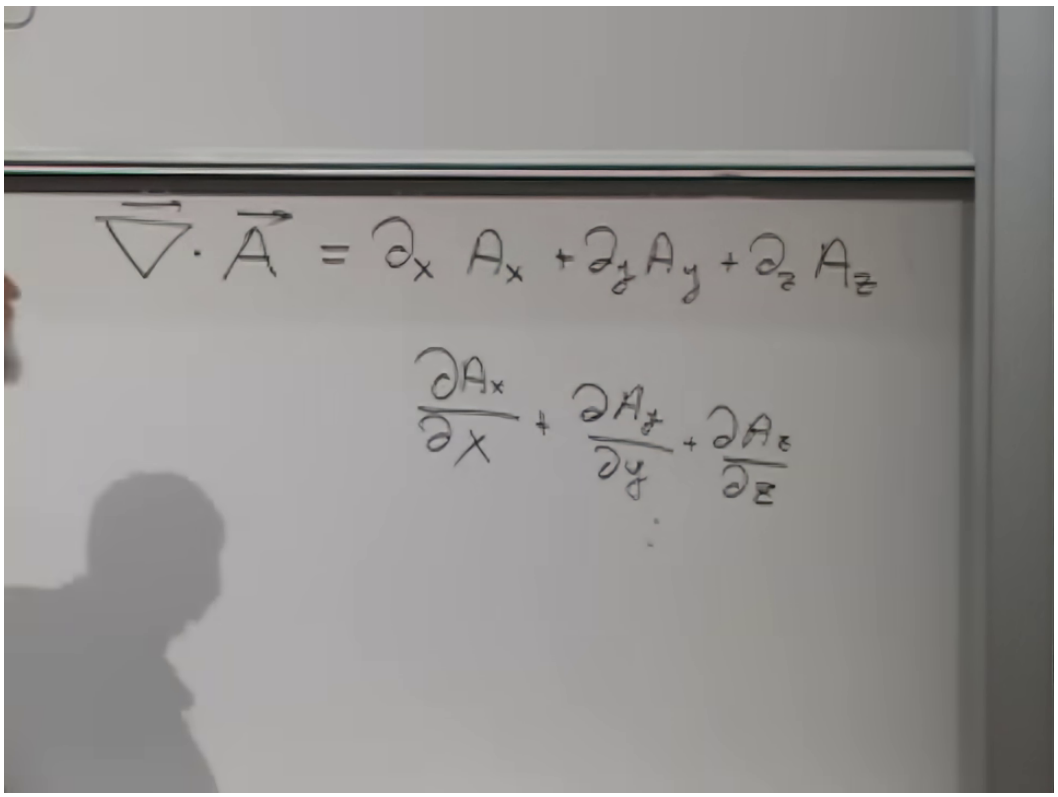


Figure 1.5: The board formula for divergence. The key distinction is between the vector field $\vec{A}$ and the scalar quantity $\nabla \cdot \vec{A}$.

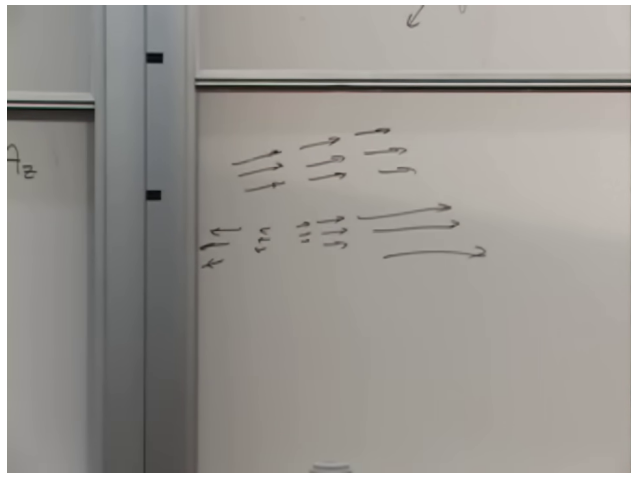


Figure 1.6: Schematic field used in the lecture to explain local divergence. The point is to compare the amount of flow entering and leaving a small control region.

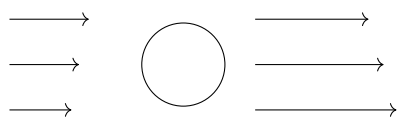


Figure 1.7: Reconstruction of the control-region picture described in the lecture. More field leaves the small circle on the right than enters from the left, so the region has positive divergence.

The board below preserves the lecture's definition.

In cleaned notation we write

$$\nabla \cdot \vec{A} = \partial_x A_x + \partial_y A_y + \partial_z A_z = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}. \tag{1.24}$$

The lecture is careful about the point that this is a scalar, not a vector. The arrows drawn on the other board are the field $\vec{A}$ itself. The divergence is the scalar quantity extracted from the spatial derivatives of its components.

To make this intuitive, the lecture turns to water flow. We imagine a shallow lake of fixed depth, or more generally an incompressible fluid whose density is fixed. Water pumped in from below creates a local source, so the velocity field spreads outward and has positive divergence. Water removed from below creates a sink, so the field converges inward and has negative divergence. The lecture later notes that one could complicate the analogy by allowing compression or thinning, but for pedagogy it keeps the picture simple: fixed depth, fixed density, visible source and sink language.

The accompanying arrow-field sketch is preserved below.

The second tool is Gauss's theorem, which ties local source strength to total flux through a boundary.

Theorem 1.3 (Gauss's theorem)**.** *For a vector field $\vec{A}$,*

$$\int_V (\nabla \cdot \vec{A})\, dV = \oint_{\partial V} A_\perp \, d\sigma = \oint_{\partial V} \vec{A} \cdot \hat{n} \, d\sigma. \tag{1.25}$$

The meaning is direct. The integral on the left adds up all sources and sinks inside the region V. The integral on the right adds up the net outward flow through the boundary ∂V. Only the normal component matters; motion tangent to the boundary slides along it and contributes no outward flux.

This is the theorem the lecture has been building toward. We now apply it to a spherically symmetric source of divergence concentrated inside some finite sphere. Let the total enclosed divergence be

$$Q = \int_V (\nabla \cdot \vec{A})\, dV. \tag{1.26}$$

If we surround that source by a larger sphere of radius r, then Q is independent of the radius of the larger sphere, provided only that the larger sphere encloses the whole source region.

By spherical symmetry, the field outside must be radial and its magnitude must be constant over the Gaussian sphere. Therefore the flux integral collapses to

$$Q = \oint_{\partial V} A_\perp \, d\sigma = A(r) \oint_{\partial V} d\sigma \tag{1.27}$$

$$= 4\pi r^2 \, A(r). \tag{1.28}$$

Hence

$$A(r) = \frac{Q}{4\pi r^2}, \qquad \vec{A}(r) = \frac{Q}{4\pi r^2}\hat{r}. \tag{1.29}$$

This is the lecture's clean derivation of a $1/r^2$ field from spherical symmetry and Gauss's theorem. It is here that Susskind briefly says "four-thirds πr^2" and immediately corrects it. Only the corrected surface area $4\pi r^2$ belongs in the notes.

Now gravity reenters. A positive source in the water analogy gives an outward field. Real Newtonian gravity points inward. So the gravitational analog is a negative source, or a convergence. For a point mass M,

$$\vec{g}(r) = -\frac{GM}{r^2}\hat{r}. \tag{1.30}$$

The lecture emphasizes that this is only a mathematical analogy. Space is not literally a fluid. But the analogy is useful because it explains why a point source naturally produces a $1/r^2$ field.

This brings us to Newton's theorem.

Proposition 1.4 (Newton's theorem)**.** *Outside a spherically symmetric mass distribution, the gravitational field is exactly the same as that of a point mass with the same total mass concentrated at the center.*

This is not a small convenience. It is what makes planetary gravity tractable. The exterior field of a spherical planet does not care whether the mass is concentrated at the center, spread out smoothly, denser in the middle, or denser near the surface. Outside the body, only the total mass matters.

1.7.1 Question & Answer

Why is the field outside a spherical body point-mass-like, while the field inside a spherical shell vanishes?

Outside the body, the answer is Gauss plus symmetry. A Gaussian sphere enclosing the entire spherically symmetric source sees only the total enclosed mass. The flux is therefore fixed. Since the field has the same magnitude everywhere on that sphere and points radially, the only possible exterior field is the $1/r^2$ field of a point mass with that same total mass.

Inside a hollow spherical shell, the answer is sharper still. A Gaussian surface entirely inside the shell encloses no source at all. Therefore the enclosed divergence is zero, and the net flux through

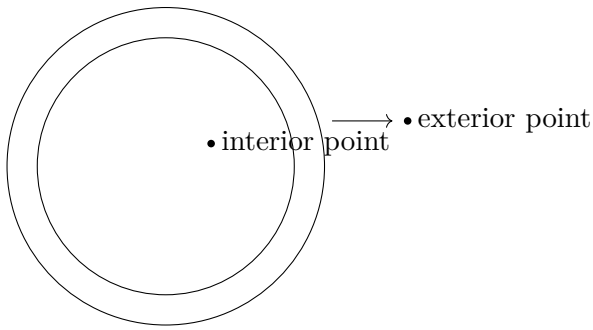


Figure 1.8: A spherical shell. Outside the shell, the field is the same as that of a point mass at the center. Inside the shell, the field vanishes.

the Gaussian surface is zero. But by symmetry there is no distinguished interior direction. The only field compatible with zero enclosed source and full spherical symmetry is

$$\vec{g}_{\text{inside shell}} = 0.$$

The lecture also insists on the corresponding limitation. This is an exterior theorem. Once we go inside a gravitating body, we are asking a different question. In the mine shaft thought experiment the field near the center of the Earth is certainly not what it would be if the entire Earth's mass were concentrated at the center. The exterior theorem does not survive unchanged at interior points.

Susskind is right to linger here. This theorem is not just elegant; it is the remarkable piece of luck that made Newtonian celestial mechanics workable.

1.8 Summary

The lecture begins with the most elementary Newtonian equation and ends with the shell theorem, but the movement is carefully staged. First we learn why gravity is special: its force is proportional to the same mass that appears in Newton's law, so in a uniform field the mass cancels and free fall becomes universal. Then we leave the flat-Earth approximation, write Newton's inverse-square law, sharpen it into vector form, and generalize it to many bodies. The motion of a test particle still ignores its own mass, which leads naturally to the gravitational field.

Only then do we stop and repair the earlier slogan. A real gravitational field varies from place to place, so large freely falling bodies are stretched and squeezed by tidal forces. After the break the lecture introduces divergence and Gauss's theorem, and these become the machinery that explains why spherical symmetry produces a $1/r^2$ law and why spherical shells are so remarkable.

That is the real payoff of Lecture 1. Already within Newtonian gravity, the mathematics begins to organize itself in a way that points beyond force law and toward geometry.

Chapter 2

Newtonian Gravity, Gauss's Law, and the Equivalence Principle

These notes follow Leonard Susskind's second lecture on general relativity in the order in which the argument is actually built. The lecture opens with a short detour about dark energy, then deliberately returns to Newtonian gravity, sharpens that old physics into field language, and only then turns toward Einstein. Where the blackboard itself carries part of the mathematical story, we keep the original lecture frames visible as documentary evidence and place the cleaned equations or diagrams nearby. Credit for the lecture belongs to Leonard Susskind; the present curation also owes explicit credit to LazyingArt LLC and Video2Book.

2.1 Dark Energy as a Prelude, Not the Main Road

The lecture begins by answering a worry that had been left hanging over from earlier exchanges: if dark energy acts like a repulsive component of gravity, why does it not tear ordinary bound systems apart? Susskind does not begin with cosmological field equations. He begins with a spring, because for the limited point he wants to make, that is the cleanest picture.

For an ordinary restoring force we may write, schematically,

$$F(r) \sim -kr. \tag{2.1}$$

A positive cosmological constant is then modeled, at this opening level, as a tiny repulsive correction of the same distance-dependent type,

$$F(r) \sim -kr + \lambda r. \tag{2.2}$$

The point is not that an atom literally is a spring. The point is that if a bound system already has a strong restoring force, then a very small repulsive term proportional to distance merely shifts the equilibrium position slightly. It weakens the binding a little; it does not destroy it.

That is the practical content of the opening discussion. The coefficient λ is so small that its effect is negligible on laboratory scales, negligible on the scale of the solar system, and still tiny compared with ordinary binding effects inside galaxies. Only on cosmological scales does such a term compete with anything else.

Susskind then sharpens the contrast by mentioning the "Big Rip" scenario. The ordinary cosmological constant does not do this. The rip would require something more violent: effectively, that the repulsive coefficient grow with time. He treats that not as sober mainstream physics, but as a bad speculative idea.

2.1.1 Question & Answer

Question. Why does dark energy not tear atoms, solar systems, or galaxies apart?

Answer. Because in the lecture's opening model the repulsive contribution is fantastically small. In a bound system it produces, to leading order, only a slight shift of the equilibrium configuration. The atom becomes a tiny bit larger; the orbit becomes a tiny bit more spread out. The "Big Rip" picture would require the repulsive effect to grow with time and eventually overwhelm binding forces. Susskind explicitly treats that as a different and much less credible story.

With the conceptual debris cleared away, the lecture returns to where it meant to go all along: we still need a little more Newton before we can earn Einstein.

2.2 A Little Formalism: Del, Fields, and the Gravitational Acceleration

Susskind introduces this next step with exactly the right tone: first a little mathematics, though really it is only formalism, a notation for organizing equations. The operator

$$\nabla = (\partial_x, \partial_y, \partial_z) \tag{2.3}$$

behaves symbolically like a vector, even though it is really a differential operator.

If $\phi(x, y, z)$ is a scalar field, then applying ∇ produces a vector field,

$$\nabla\phi = (\partial_x\phi,\, \partial_y\phi,\, \partial_z\phi)\,. \tag{2.4}$$

If instead $\vec{v}(x, y, z)$ is a vector field, then the divergence is the scalar

$$\nabla \cdot \vec{v} = \partial_x v_x + \partial_y v_y + \partial_z v_z. \tag{2.5}$$

We should keep the bookkeeping completely straight here. The lecture contains a fleeting verbal slip, but the mathematics is unambiguous: the gradient of a scalar is a vector, while the divergence of a vector is a scalar.

A scalar field is just a scalar quantity depending on position. A vector field is a vector quantity depending on position. The Newtonian gravitational field is such a vector field.

Now Susskind defines it operationally. Take a distribution of source masses m_i at positions $\vec{x}_i$. Introduce a tiny test mass, release it at the field point $\vec{x}$, and observe its acceleration. That acceleration is the gravitational field:

$$\vec{A}(\vec{x}) = \text{acceleration of a test mass released at } \vec{x}. \tag{2.6}$$

There is a brief and telling hesitation over the sign convention. That hesitation should not be erased, because it shows exactly what must be fixed. We want the field to point *toward* the source mass. Therefore we choose

$$\vec{R}_i := \vec{x}_i - \vec{x}, \tag{2.7}$$

so that $\vec{R}_i$ points from the test mass toward the ith source. With that choice the Newtonian field is

$$\vec{A}(\vec{x}) = G\sum_i m_i \frac{\vec{R}_i}{R_i^3} = G\sum_i m_i \frac{\hat{R}_i}{R_i^2}, \qquad \hat{R}_i = \frac{\vec{R}_i}{R_i}. \tag{2.8}$$

That is simply the inverse-square law rewritten as a field at a point. The magnitude scales like $1/R_i^2$; the direction is carried by $\vec{R}_i$.

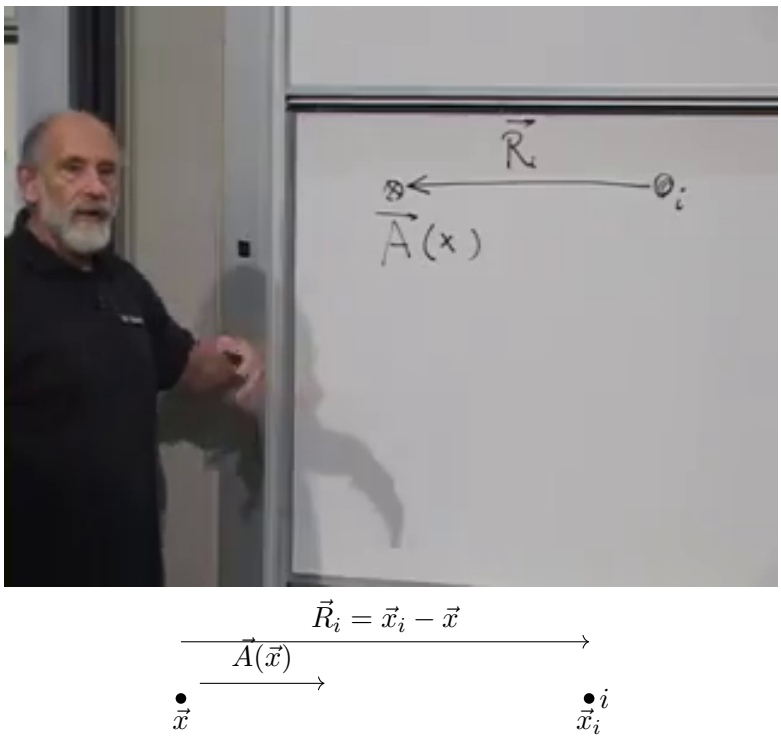


Figure 2.1: Field vector and source-point displacement. The board frame is kept because it is the clearest documentary evidence for the source-point/field-point geometry, while the redraw fixes the final sign convention chosen in the lecture.

Up to this point the field has been assembled one source at a time. The next move is the one Susskind explicitly advertises: there is a neat way to summarize all this.

2.3 Gauss's Law as Physics, Gauss's Theorem as Mathematics

To pass from discrete particles to a field equation, we introduce the mass density. If a small volume ΔV contains a small mass Δm, then

$$\rho = \frac{\Delta m}{\Delta V}. \tag{2.9}$$

In the continuum picture $\rho(\vec{x})$ is itself a scalar field.

Now comes the summary of Newtonian gravity:

$$\nabla \cdot \vec{A} = -4\pi G\rho. \tag{2.10}$$

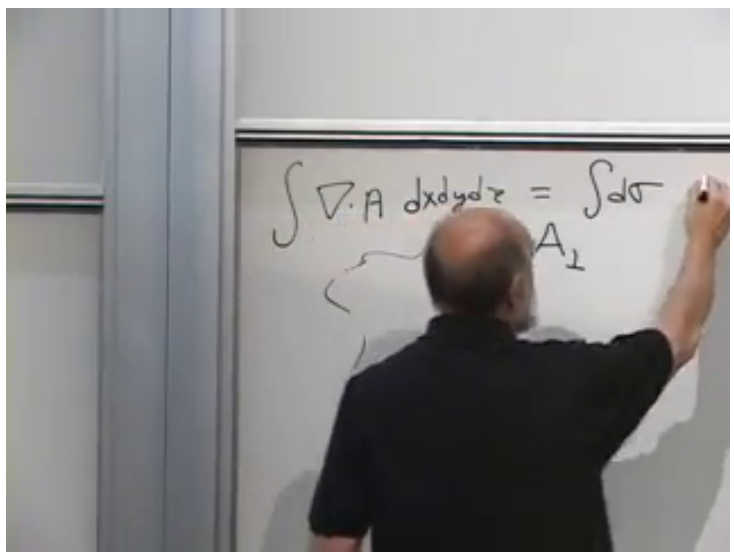
This is Gauss's law for gravity. The minus sign records the fact that the field points inward toward positive mass. The field converges. A convergence is a negative divergence.

Susskind is especially careful here to separate two statements that are often blurred together. Gauss's law is physics. It is a law of nature, equivalent to Newtonian gravity. Gauss's theorem is mathematics. It says that for any sufficiently nice region V with boundary ∂V,

$$\int_V (\nabla \cdot \vec{A})\, dV = \oint_{\partial V} A_\perp \, d\sigma. \tag{2.11}$$

On the left we integrate the divergence throughout the volume. On the right we integrate the component of the field normal to the boundary. The theorem translates volume divergence into boundary flux.

The blackboard frame for this moment is only partially legible, but the intent is clear enough to justify writing the standard form.



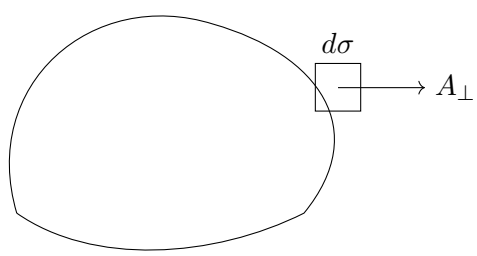


Figure 2.2: Surface-flux term in Gauss's theorem. The lecture frame shows the half-written theorem and the boundary patch labeled $A_\perp$; the redraw makes explicit that the theorem uses the normal component through the surface element $d\sigma$.

This is still abstract. The lecture does not linger there. Susskind immediately says, in effect, let us put them together and see what they do.

2.4 Putting Them Together: Exterior Fields and the Shell Theorem

Suppose the mass distribution is spherically symmetric, or isotropic, about a center. Then we surround it by a concentric Gaussian sphere of radius r. Now the theorem and the law meet each other.

From Gauss's law,

$$\nabla \cdot \vec{A} = -4\pi G\rho, \tag{2.12}$$

so inside the volume integral we have

$$\int_V (\nabla \cdot \vec{A})\, dV = -4\pi G \int_V \rho\, dV. \tag{2.13}$$

But the integral of ρ over the volume is simply the enclosed mass,

$$\int_V \rho\, dV = M_{\text{enc}}. \tag{2.14}$$

Meanwhile, on the surface side, symmetry tells us something equally important. The field must be radial, and its radial component has the same value at every point of the Gaussian sphere. Therefore

$$\oint_{\partial V} A_\perp\, d\sigma = 4\pi r^2 A_r. \tag{2.15}$$

Putting these together gives

$$4\pi r^2 A_r = -4\pi G M_{\text{enc}}, \tag{2.16}$$

and therefore

$$A_r = -\frac{GM_{\text{enc}}}{r^2}. \tag{2.17}$$

If the sphere encloses the whole body, then $M_{\text{enc}} = M$, and we recover the familiar inverse-square field

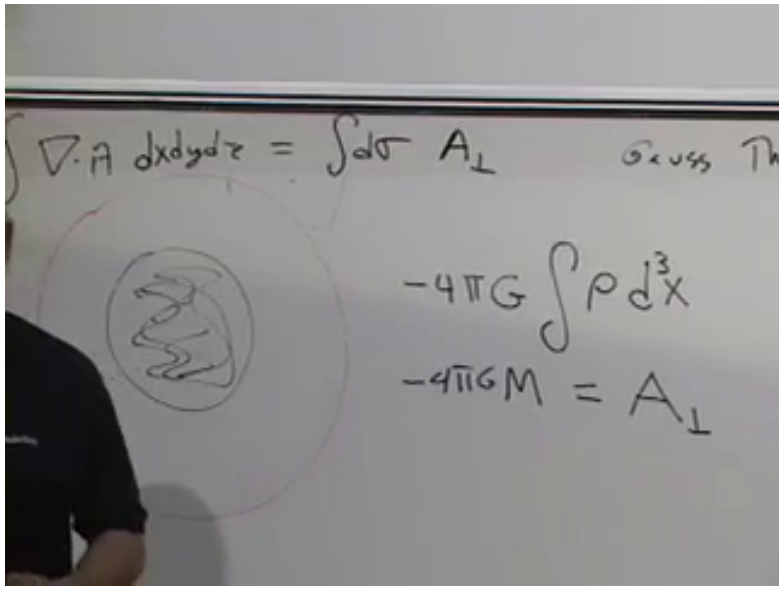
$$A_r = -\frac{GM}{r^2}. \tag{2.18}$$

This is also the point in the lecture where Susskind remarks on the appearance of the factor 4π. In this normalization it is chosen so that the surface area of the sphere cancels cleanly. One could normalize the equations differently, but this choice makes spherical problems fall open at once.

What has happened is more than a recovery of Newton's formula. We have also learned something new in this language: the exterior field of any spherically symmetric mass distribution is the same as the field of a point mass at the center.

2.4.1 Question & Answer

Question. Why does only the mass inside the Gaussian sphere matter?



Gaussian sphere

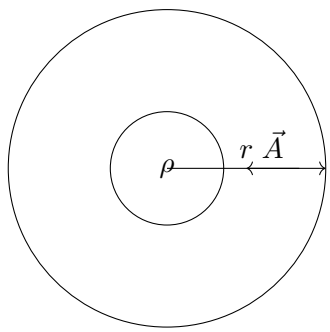


Figure 2.3: Gauss law with enclosed mass. The lecture frame shows the shift from the abstract flux theorem to the gravitational mass integral; the redraw isolates the concentric spherical geometry used in the derivation.

Answer. Because the flux through the Gaussian sphere is fixed by the integral of the density *inside* that sphere. Once spherical symmetry is assumed, the flux determines the radial field on the boundary. Matter outside contributes zero net field there. This is the shell theorem in flux form.

That answer has a remarkable corollary. If the entire mass were placed on a thin spherical shell, then the field inside the shell would vanish identically. Outside, the field would be exactly the same as that of a point mass at the center. Susskind emphasizes that Newton knew this and proved it geometrically, without calculus. In these notes we see it fall directly out of Gauss's law.

2.5 While We Are At It: Inside the Earth

At this point Susskind turns, in the same spirit, to a question from the previous meeting: what does the gravitational field look like *inside* the Earth? The same formalism answers it almost for free.

We again choose a Gaussian sphere, but now the sphere lies inside the Earth. The flux side is

unchanged:

$$\oint_{\partial V} A_\perp \, d\sigma = 4\pi r^2 A_r, \tag{2.19}$$

where r is now the radius of the *inner* Gaussian sphere. What changes is the mass enclosed.

To make the calculation explicit, Susskind adopts a simplified Earth: uniform density ρ throughout the interior, abruptly dropping to zero at the surface. Then

$$M_{\text{enc}}(r) = \frac{4\pi}{3} r^3 \rho. \tag{2.20}$$

Substituting this into Gauss's law gives

$$4\pi r^2 A_r = -4\pi G M_{\text{enc}}(r), \tag{2.21}$$

$$4\pi r^2 A_r = -4\pi G \left(\frac{4\pi}{3} r^3 \rho \right). \tag{2.22}$$

The board algebra in the lecture becomes momentarily unstable here, and the audience catches a missing factor of 4π. The corrected result, which is the one we should preserve, is

$$A_r = -\frac{4\pi G \rho}{3} \, r. \tag{2.23}$$

This also matches the exterior formula at the surface $r = R$ once we use

$$M = \frac{4\pi}{3} R^3 \rho. \tag{2.24}$$

Now the lecture cashes out the result physically. Since $\vec{A}$ is an acceleration field, multiplying by the mass m of the test particle gives

$$m\ddot{r} = -\frac{4\pi G \rho}{3} \, m r. \tag{2.25}$$

This is the equation of a harmonic oscillator. Inside the idealized uniform Earth, gravity becomes a linear restoring force.

That is a lovely Newtonian payoff. A particle displaced from the center is pulled back exactly as though attached to a spring. In the tunnel-through-the-Earth thought experiment, the particle would oscillate through the planet.

2.5.1 Question & Answer

Question. How can the field be linear in r if the enclosed mass grows like r^3?

Answer. Because Gauss's law fixes the *flux*, not the field directly. The flux contains an area factor $4\pi r^2$. Thus the scaling is

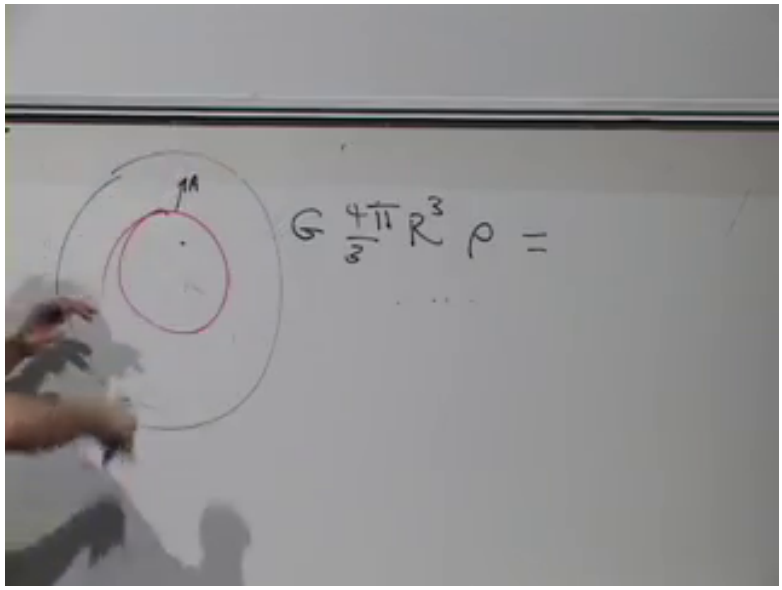
$$A_r \, r^2 \sim r^3, \tag{2.26}$$

so only one power of r survives:

$$A_r \sim r. \tag{2.27}$$

This is exactly the local conceptual snag that surfaces in the classroom, and it is useful because it makes visible the role of the Gaussian sphere itself.

The lecture briefly announces gravitational potential here, but then defers it. That deferral is important. The Newtonian recap has reached the point it needed to reach, and Susskind is ready to say so: this has been old physics. Now we move to the new idea.



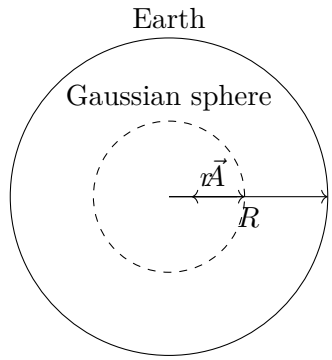


Figure 2.4: Concentric spheres and enclosed-mass factor. The preserved lecture frame supplies the documentary evidence for the inner Gaussian sphere and the visible $\frac{4\pi}{3}r^3\rho$ factor, while the redraw gives the cleaned geometry for the corrected derivation.

2.6 The Equivalence Principle as Elevator Intuition and Coordinate Transformation

The equivalence principle first appears in the lecture in the form everybody remembers: the elevator. Suppose we have a cabin accelerating horizontally with acceleration g. A passenger releases an object inside. In the inertial frame outside the elevator, the object simply continues its motion. But to the observer riding inside, the object accelerates backward toward the floor of the elevator with magnitude g. It looks like gravity, sounds like gravity, smells like gravity.

Susskind insists that this alone would not make Einstein Einstein. The real step is not merely to say the words. The real step is to use them. But before he uses them, he rebuilds the elevator mathematically.

He chooses the horizontal coordinate x precisely because he wants to draw an x–t diagram. First consider uniform velocity. If the floor of the elevator moves with constant speed v, then the passenger's coordinate relative to the floor is

$$x' = x - vt, \qquad t' = t. \tag{2.28}$$

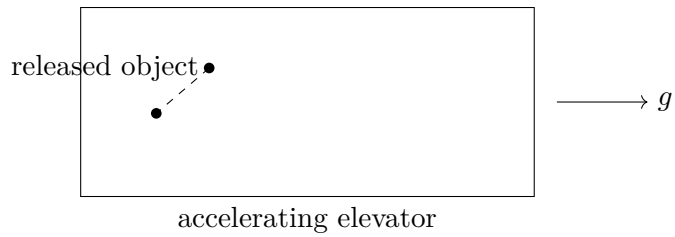


Figure 2.5: Horizontal elevator version of the equivalence principle. In the passenger's frame, the released object accelerates backward toward the floor exactly as if a uniform gravitational field were present.

This is still Newtonian physics: all observers use the same time. If an object is at rest in the inertial frame, then $x =$ constant and hence

$$\dot{x}' = -v, \qquad \ddot{x}' = 0. \tag{2.29}$$

So a uniformly moving frame creates no fictitious gravitational field.

Now let the elevator accelerate uniformly. Its floor then follows

$$x_{\text{floor}}(t) = \frac{1}{2}gt^2, \tag{2.30}$$

and the corresponding transformed coordinate is

$$x' = x - \frac{1}{2}gt^2. \tag{2.31}$$

For an object at rest in the inertial frame, we differentiate once and then again:

$$\dot{x}' = -gt, \qquad \ddot{x}' = -g. \tag{2.32}$$

This is the mathematical statement of the elevator argument. In the accelerated frame, freely moving bodies behave as though there were a uniform gravitational field.

This is not yet full relativity. Susskind says so directly. But it already gives the conceptual bridge he wants: acceleration corresponds to a change to curvilinear coordinates in space-time. That is why the discussion is heading toward geometry.

2.7 Einstein's Move: Light, Locality, and the Metric

Now comes the genuinely Einsteinian use of the principle. If gravity and acceleration are equivalent only for falling stones, then the principle is not deep. Einstein's claim is stronger: the laws of physics in a gravitational field are the same as the laws of physics in an accelerated frame. Therefore the principle must apply to light as well.

Imagine an elevator standing still. A light beam sent horizontally across it travels in a straight line and hits the opposite wall directly across. Now imagine the elevator accelerating upward. The light beam still begins horizontally, but by the time it reaches the far wall the elevator has moved upward. So in the elevator frame the light lands lower than before. Its path is curved. In fact, in this approximation, it is parabolic.

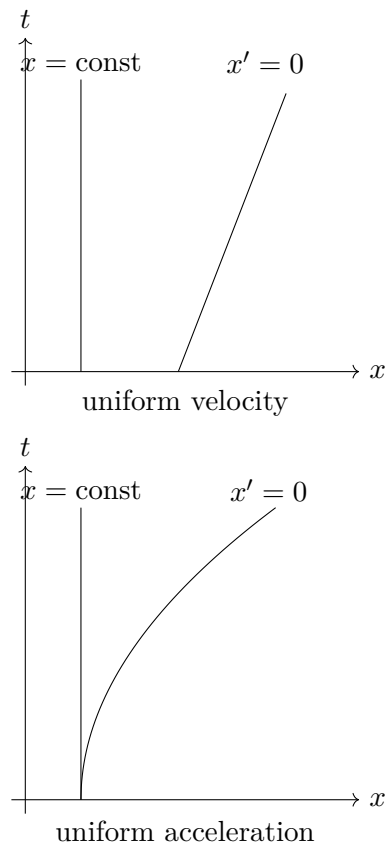


Figure 2.6: x–t sketches for the moving and accelerating elevator. The accelerating floor follows a parabola, and the transformed coordinate produces the effective field $\ddot{x}' = -g$.

That simple argument leads to Einstein's early estimate for light bending by the Sun. The lecture keeps the estimate rough and historical, and the notes should do the same. Consider a light ray that just skims the solar surface. Near closest approach, take the downward acceleration to be roughly the surface gravity,

$$g_\odot \approx \frac{GM}{R^2}. \tag{2.33}$$

The time spent in the region where this matters is estimated by the light-crossing time of about two solar radii:

$$\Delta t \approx \frac{2R}{c}. \tag{2.34}$$

Hence the downward component of the velocity picked up by the ray is

$$\Delta v_\perp \approx g_\odot \Delta t \approx \frac{2GM}{Rc}. \tag{2.35}$$

For a small deflection angle,

$$\theta \approx \frac{\Delta v_\perp}{c} \approx \frac{2GM}{Rc^2}. \tag{2.36}$$

This is Einstein before the full apparatus of general relativity. Susskind keeps the historical caution in view: the final relativistic answer differs by a factor of two. The point of the lecture here is not precision but the conceptual use of the equivalence principle in a place where nobody had used it before.

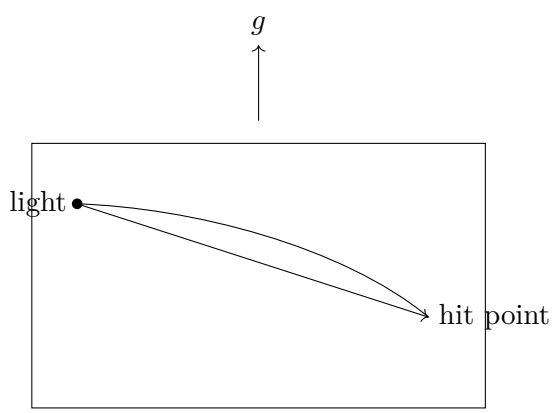


Figure 2.7: Light in an upward-accelerating elevator. This is the elevator version of the statement that light falls in a gravitational field.

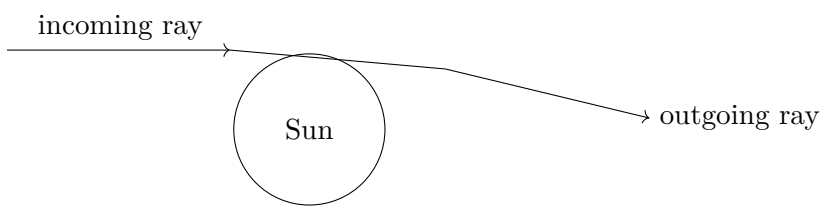


Figure 2.8: Crude sun-skimming-ray estimate. The lecture treats the deflection qualitatively and then estimates it by combining surface gravity with a light-crossing time.

2.7.1 Question & Answer

Question. Can free fall completely eliminate gravity, or only locally?

Answer. Only locally. A sufficiently small elevator, observed for a sufficiently short time, can replace the Earth's field by a single accelerating frame. But the real Earth's field is not uniform. Its direction varies from place to place, and its magnitude varies with distance. A large enough freely falling body notices this immediately: its feet are pulled more strongly than its head, and neighboring free-fall lines converge sideways. Those are tidal forces. They are the obstruction to eliminating gravity globally.

That is the limiting correction Susskind insists on before the lecture ends. Gravity can be transformed away only in a local neighborhood. The real, invariant content of a gravitational field lies in what survives that attempt, namely tidal effects.

He then steps back from mechanics and turns to the blackboard itself. How does one describe the geometry of a flat two-dimensional surface? In rectangular Cartesian coordinates, for neighboring points,

$$ds^2 = dx^2 + dy^2. \tag{2.37}$$

But in curvilinear coordinates on the same flat board, the distance between neighboring points is more generally

$$ds^2 = g_{xx}\,dx^2 + 2g_{xy}\,dx\,dy + g_{yy}\,dy^2. \tag{2.38}$$

The cross term tells us that the coordinate lines need not be perpendicular. The position dependence

of the coefficients tells us that the coordinate mesh can stretch, squeeze, and twist from place to place. These functions are the metric.

Now the final bridge of the lecture can be stated cleanly. Curvature is the obstruction to finding coordinates that globally flatten the geometry. Tidal forces are the obstruction to finding coordinates that globally eliminate the gravitational field. The connection is not accidental. It is the first glimpse of general relativity as geometry.

2.8 Summary

Lecture 2 begins with a seemingly peripheral question about dark energy, but that opening detour performs useful work: it clears away confusion before the main argument returns to Newton. The Newtonian part of the lecture is then rebuilt in a more modern field language. We define the gravitational field as the acceleration of a test mass, summarize the many-body inverse-square law by

$$\nabla \cdot \vec{A} = -4\pi G\rho,$$

and combine Gauss's law with Gauss's theorem to recover both the exterior inverse-square field and the linear interior field of a uniform-density Earth.

Only after that long Newtonian runway does Einstein's idea become sharp. An accelerated elevator produces an effective gravitational field; the transformed coordinate satisfies $\ddot{x}' = -g$; and once one takes that principle seriously for all physics, light itself must bend. But the principle is only local. Tidal forces remain. That is why the lecture ends with the metric and curvature: not yet with the full theory, but with the right obstruction, the right bridge, and the right sense that gravity is what cannot finally be transformed away.

Chapter 3

Vertical Light Rays, Tensors, and the Metric

This lecture begins by filling in a loose end from the equivalence principle and ends by introducing the metric tensor. The path matters. We first ask what happens to a vertical light ray in an accelerating elevator or in a gravitational field, then we sharpen that into a question about frequency shift, then we move to free fall and discover what can and cannot be transformed away. Only after that obstruction is clearly in view do we build the tensor language that will let us describe curved geometry carefully. These notes follow Leonard Susskind's lecture closely; where blackboard frames are kept below, they serve only as documentary support for equations and diagrams, in the spirit of the LazyingArt LLC and Video2Book curation, while the transcript remains the primary guide to the argument.

3.1 Vertical Light Rays, Doppler Shift, and the Equivalence Principle

Susskind begins by returning to a question that was left hanging: not a light ray crossing the elevator, but a light ray traveling vertically. If we shoot an ordinary bullet upward, gravity slows it down; by the time it reaches the ceiling it has less kinetic energy than when it left the floor. That is the familiar Newtonian story.

But light is not an ordinary bullet. Everybody agrees on the speed of light. So when we ask what gravity does to a vertical light ray, the useful question is not whether the speed changes, but what the detector reads when the light arrives. The answer is that the color changes, or in more controlled language, the frequency changes.

Susskind chooses to discuss the downward beam first. Suppose the elevator is momentarily at rest when the beam is emitted from the roof. By the time the beam reaches the floor, the elevator has acquired an upward velocity. The detector on the floor is therefore intercepting the beam while moving relative to the source, and the received light is Doppler shifted. In the accelerating elevator the downward beam is detected slightly blue shifted. By the equivalence principle, the same conclusion applies to a beam fired downward in a gravitational field.

This is not presented as a trick of language. The same effect has been tested near the Earth. Reverse

the experiment and the effect reverses with it: a beam sent upward is detected red shifted. Solar spectral lines are correspondingly shifted, and for sufficiently compact objects the effect becomes very large. Susskind emphasizes that the relevant issue is not just size, but compactness; what matters is how hard it is for light to climb out.

3.1.1 Question & Answer

Question. Why do the Doppler shifts not simply cancel?

Answer. They would cancel if the source and the detector differed only by a constant velocity. In that case one shift would be attributed to emission, the opposite one to detection, and the two would undo each other. But here the elevator is accelerating. When the beam is fired, the source is at rest in the chosen frame. By the time the beam arrives, the detector has a different velocity. It is the accelerated motion that makes the later detector velocity differ from the earlier emitter velocity.

Susskind does not write the numerical estimate on the board, but he explicitly assigns it as an exercise. If the height of the elevator is h, then the light travel time is approximately

$$t \approx \frac{h}{c}. \tag{3.1}$$

In that time the detector acquires speed

$$v \approx gt \approx \frac{gh}{c}, \tag{3.2}$$

and therefore the fractional frequency shift is of order

$$\frac{\Delta\nu}{\nu} \sim \frac{v}{c} \sim \frac{gh}{c^2}. \tag{3.3}$$

For the downward beam the shift is blue; for the upward beam the sign is reversed.

Remark 3.1. The estimate $\Delta\nu/\nu \sim gh/c^2$ is a cautious reconstruction of the exercise Susskind assigns. It is not one of the formulas visibly written on the board in the selected frames.

The lecture broadens the point immediately after this. Near the Earth the shift is small; for the Sun it is still small but observable; for very compact stars it is larger; near a black hole it becomes enormous. So the opening thought experiment is not just pedagogical. It is already a real physical prediction of the equivalence principle.

3.2 Free Fall, Local Cancellation, and Tidal Obstruction

The next pivot is crucial. An accelerated frame is not something one can fail to notice. If the elevator is being driven upward, one can certainly tell that one is not inertial; one may say either that one is accelerating or that one feels gravity, but one does not mistake the frame for free space. The subtle case is the freely falling elevator.

So we repeat the same experiment in a real gravitational field, but now with the elevator in free fall. Once again Susskind takes a downward beam. There are now two effects to keep track of. The gravitational field by itself would shift the beam one way. But by the time the beam reaches the

floor, the floor is itself moving, and that induces an ordinary Doppler shift the other way. The equivalence principle says that in a freely falling elevator these two effects cancel exactly, apart from tidal forces.

We may summarize the local claim as

$$\Delta\nu_{\text{grav}} + \Delta\nu_{\text{Doppler}} = 0, \tag{3.4}$$

with the understanding that this is a local statement. Susskind is careful here: if one insists on doing the calculation exactly, one must use special relativity. A purely nonrelativistic treatment leaves a spurious residual at higher order in v/c, whereas the relativistic treatment gives the exact cancellation demanded by the equivalence principle.

3.2.1 Question & Answer

Question. Why is there no net shift in a freely falling elevator even though the elevator is accelerating?

Answer. Because free fall is precisely the situation in which the acceleration of the laboratory compensates for the local effect of gravity. If the cancellation failed, a freely falling laboratory would be distinguishable from an inertial laboratory in empty space, and Einstein's equivalence principle would be wrong. What remains, and what cannot be removed in this way, is only the variation of gravity across the size of the elevator.

That last phrase is the real transition. Gravity near the bottom is not quite the same as gravity near the top. Those leftover effects are the tidal forces. One can remove gravity in a small enough region and for a short enough time, but one cannot remove it globally. There is an obstruction.

Before turning to mathematics, Susskind pauses to reinterpret the accelerated elevator in coordinate language. Meter sticks that are at rest in the elevator trace curved trajectories when viewed from an inertial frame. From outside, one may therefore blame the peculiar phenomena seen by the observer inside the elevator on the use of curved coordinates. In that sense, the "gravity" in the uniformly accelerated elevator can be traded for coordinate curvature.

But real gravity due to a planet or star is not exhausted by that story. A freely falling elevator removes most of the effect, not all of it. Tidal forces survive, and so there is an obstruction to removing the gravitational field by a single global coordinate transformation. That obstruction, Susskind now says explicitly, is closely related to the obstruction to removing the curvature of a curved space. That is the bridge into differential geometry.

3.3 The Minimal Calculus Toolkit for Curved Coordinates

Having reached the bridge, the lecture deliberately slows down. We are not yet doing spacetime; for the moment we are speaking only about space. The mathematics, Susskind insists, is not more exotic than multivariable calculus. But we do need two basic rules, and we need them in a notation suited to repeated coordinate changes.

Take a d-dimensional space with coordinates

$$x^1, x^2, \ldots, x^d.$$

A scalar field is simply a function $\phi(x)$, meaning a single number assigned to each point. Now move an infinitesimal amount. The coordinates change by dx^m, and the field changes by $d\phi$. The first governing formula is the total differential:

$$d\phi = \frac{\partial \phi}{\partial x^m}\, dx^m. \tag{3.5}$$

This is only the ordinary statement that the change in ϕ is the sum of its directional changes along each coordinate axis.

The next step is the Einstein summation convention. Instead of writing a summation symbol every time, we agree that a repeated index means a sum over that index. Thus

$$\frac{\partial \phi}{\partial x^m}\, dx^m$$

means the sum over $m = 1, \ldots, d$. The repeated index is a dummy index: we could have called it m, n, or r; the letter does not matter, provided it does not collide with a free index elsewhere in the same equation.

Coordinate assumptions. The lecture is explicit that a coordinate system is not arbitrary scribbling. Each point has a unique set of coordinates, and each allowed coordinate map must be one-to-one, continuous, differentiable, and smooth enough for the derivatives we need. The d in $x^1, \ldots, x^d$ is the dimension of the space, so changing coordinates does not change the number of coordinates. The lecture also pauses briefly to say that all the quantities being multiplied here are ordinary commuting numbers, and that the symbol d indicates a first-order infinitesimal change, not a finite difference Δ.

Now introduce a second coordinate system $y^n = y^n(x)$. The second governing formula is the chain rule for coordinate change:

$$\frac{\partial \phi}{\partial y^n} = \frac{\partial \phi}{\partial x^m}\frac{\partial x^m}{\partial y^n}. \tag{3.6}$$

Susskind spends time unpacking what this means for a fixed n, precisely because it is going to be reused over and over. One first differentiates ϕ with respect to the x's, then differentiates the x's with respect to the y's, and then sums over the repeated index. These two formulas, the total differential and the chain rule, are the small tools from which the tensor machinery will be built.

3.4 Scalars, Vectors, and Coordinate Dependence of Components

Once those rules are in place, the lecture returns to geometry. The key point is that a geometric object is not the same thing as the list of its components. If we overlay one coordinate grid on another, then the same point and the same vector can be described in either system, but the numerical components depend on the coordinates we choose.

Susskind begins with the easiest case. A scalar has no directional content. If the x-coordinates and the y-coordinates refer to the same point, then the scalar value at that point is the same in both descriptions:

$$\phi(y) = \phi(x). \tag{3.7}$$

This is not yet mysterious. The temperature at a point does not care how we label the point.

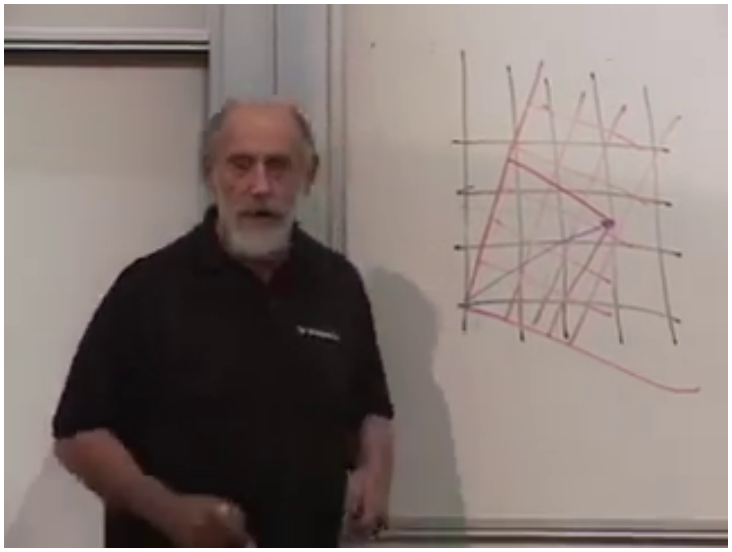


Figure 3.1: Skew coordinate grid and vector construction. The black and red coordinate families describe the same plane in different ways.

We now come to the "primordial" vector of the lecture: the infinitesimal displacement. On the board Susskind marks it by an arrow over dx, but the exact chalk styling is not essential. What matters is the idea. The displacement is a geometric object. Its components in the x-system are dx^m, and its components in the y-system are dy^n.

Because each y^n is an ordinary function of the x's, we may apply the total-differential formula to y^n itself. That gives

$$dy^n = \frac{\partial y^n}{\partial x^m}\, dx^m. \tag{3.8}$$

This is the first nontrivial transformation law in the lecture. The same vector is being described in two coordinate systems, and the components are related by the Jacobian $\partial y^n/\partial x^m$.

Definition 3.2. A contravariant vector is a collection of components V^m whose components transform according to

$$V^n_{(y)} = \frac{\partial y^n}{\partial x^m}\, V^m_{(x)}. \tag{3.9}$$

The infinitesimal displacement dx^m is the model example.

The point here is not that a vector is "made of" its components. The vector itself is geometric; the components depend on the coordinate frame. That is exactly the lesson carried by the board sketches.

3.5 From Vectors to Tensors, and from Contravariant to Covariant

Once the contravariant vector law is in hand, Susskind does not jump to an axiomatic definition of all tensors at once. He builds the hierarchy. If A^m and B^n are vectors, then the products $A^m B^n$ form a rank-two object. Each index transforms the way a vector index transforms, so each index contributes its own Jacobian factor. That is the pattern.

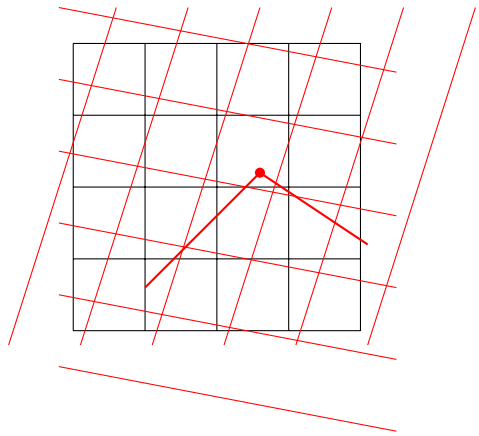


Figure 3.2: A schematic redraw of the board's point: the geometry stays fixed while the coordinate families change.

The next important object is not a displacement but a gradient. The collection $\partial\phi/\partial x^m$ has d components just as dx^m does, but it is geometrically different. It tells us how a scalar field changes along different directions. From the chain rule we read off its transformation law immediately:

$$A_n^{(y)} = \frac{\partial x^m}{\partial y^n} A_m^{(x)}. \tag{3.10}$$

An object with this transformation law is called a covariant vector. The lecture's model example is the gradient $A_m = \partial\phi/\partial x^m$.

This is also where the upstairs and downstairs notation begins to earn its keep. Contravariant indices are written upstairs, covariant indices downstairs. The notation is not yet being presented as deep geometry. At first it is a disciplined bookkeeping system. But it is a very good one: when we sum over indices, we sum an upstairs index against a downstairs index. By inspection we can often see whether an equation is well formed.

A tensor with two lower indices then transforms with one factor of $\partial x/\partial y$ for each lower index:

$$T_{mn}^{(y)} = \frac{\partial x^r}{\partial y^m}\frac{\partial x^s}{\partial y^n} T_{rs}^{(x)}. \tag{3.11}$$

The lecture insists on the pattern more than on an abstract definition. A scalar has no indices and does not change. A contravariant vector transforms with $\partial y/\partial x$. A covariant vector transforms with $\partial x/\partial y$. A tensor gets one such factor for each of its indices. That is the machinery we need in order to come to the metric.

3.6 The Metric Tensor from Pythagoras

After the break there is a deliberate reset. We now come to the metric tensor, and Susskind starts as simply as possible: with flat space. He is careful to distinguish flat *space* from flat *coordinates*. A flat space can perfectly well be described by curvilinear coordinates. What makes it flat is that there exists at least one coordinate system in which the metric takes the Cartesian form.

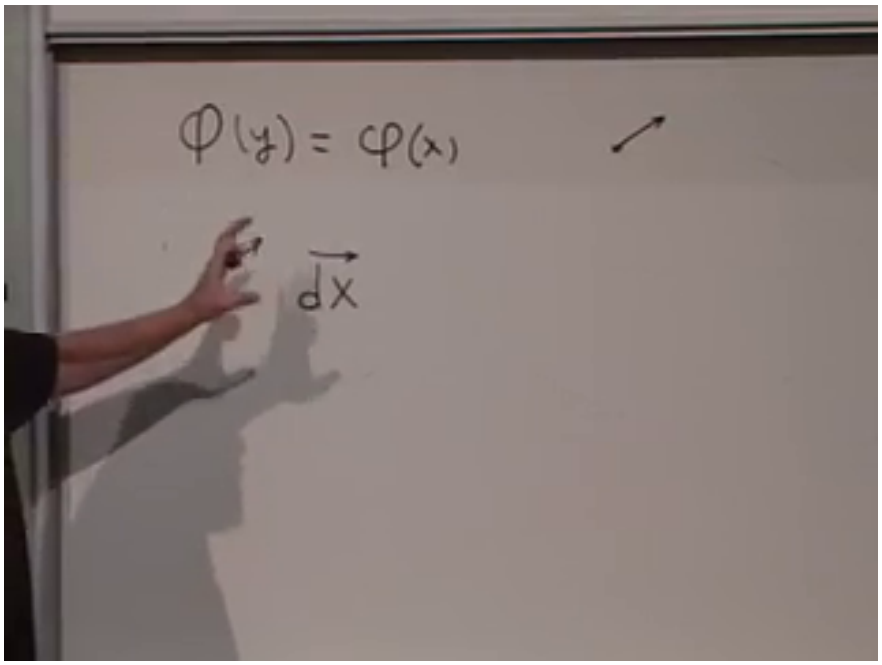


Figure 3.3: Scalar equality and free vector. The board pairs scalar invariance with the geometric character of the displacement vector.

$$\phi(y) = \phi(x)$$

Figure 3.4: A small schematic echo of the board: the same free vector may be drawn in different places without changing the geometric object.

Take an infinitesimal displacement dx. Its squared length is ds^2. In Cartesian coordinates, Pythagoras gives

$$ds^2 = (dx^1)^2 + (dx^2)^2 + \cdots . \tag{3.12}$$

At this point Susskind pauses over notation. The superscript on dx^m labels the coordinate component; it is not an exponent. By contrast, the superscript in ds^2 is a genuine square.

One can write the sum schematically as $dx^m dx^m$, but that violates the upstairs-downstairs contraction pattern introduced earlier. The cure is to insert the Kronecker delta, defined by

$$\delta_{mn} = 1 \quad \text{when } m = n, \qquad \delta_{mn} = 0 \quad \text{when } m \neq n.$$

Then the line element takes the cleaner form

$$ds^2 = \delta_{mn}\, dx^m dx^n. \tag{3.13}$$

The Kronecker delta picks out the square terms and kills the mixed terms. In Cartesian coordinates it is therefore the metric tensor.

Now suppose we change coordinates from x to y. The first step, clearly visible on the board, is

$$dx^m = \frac{\partial x^m}{\partial y^r}\, dy^r. \tag{3.14}$$

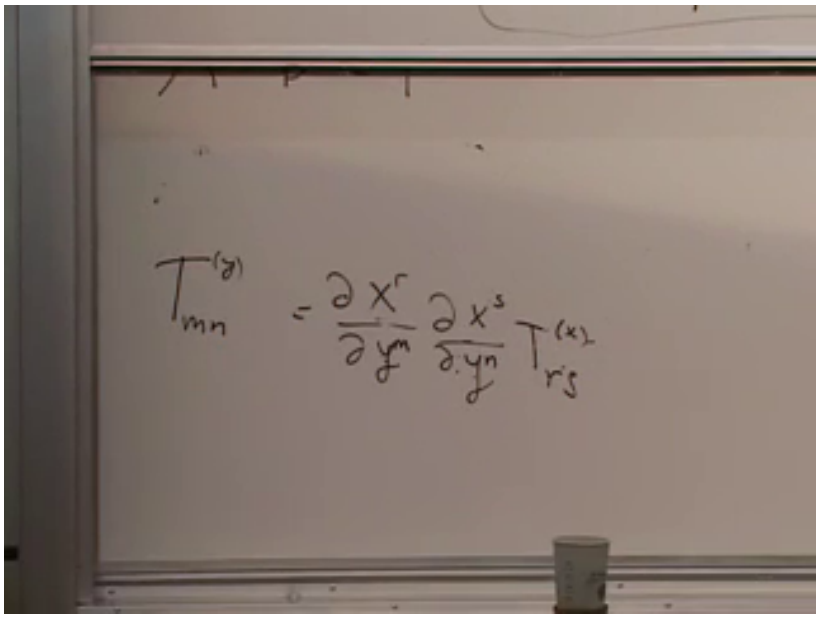


Figure 3.5: Rank-two tensor transformation law. The board shows the covariant two-index rule just before the metric tensor is introduced.

Substitute this into the line element. The board does this top to bottom, and the layout is pedagogically important: first the coordinate differential, then the substituted line element, then the identification of the new coefficient as the metric.

In clean notation, the argument is

$$ds^2 = g^{(x)}_{mn}\, dx^m dx^n \tag{3.15}$$

$$= g^{(x)}_{mn} \left(\frac{\partial x^m}{\partial y^r}\, dy^r \right) \left(\frac{\partial x^n}{\partial y^s}\, dy^s \right) \tag{3.16}$$

$$= \left(g^{(x)}_{mn} \frac{\partial x^m}{\partial y^r} \frac{\partial x^n}{\partial y^s} \right) dy^r dy^s. \tag{3.17}$$

The coefficient in parentheses is precisely the quantity that Susskind circles and names on the board, so we define

$$g^{(y)}_{rs} = g^{(x)}_{mn} \frac{\partial x^m}{\partial y^r} \frac{\partial x^n}{\partial y^s}. \tag{3.18}$$

This is the transformation law of a covariant tensor with two lower indices. The metric is therefore a covariant rank-two tensor.

From this point onward, the metric is the central geometric object. It tells us the distance between neighboring points:

$$ds^2 = g_{mn}(x)\, dx^m dx^n. \tag{3.19}$$

To know the metric is to know the local geometry.

3.7 Flatness, Curvature, and the Meaning of an Obstruction

The lecture now cashes out the word obstruction in geometrical language. Suppose we are given a general metric $g_{mn}(x)$. When does it describe flat space, and when does it describe curved space?

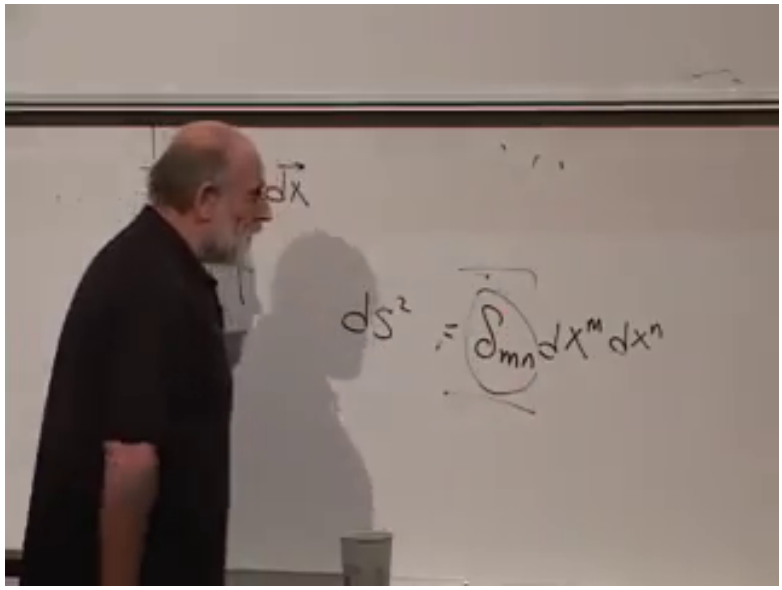


Figure 3.6: Metric tensor in Cartesian coordinates. The circled δ_{mn} marks the point at which the Cartesian metric is explicitly identified.

Susskind's answer is not a local curvature formula; it is a coordinate question.

Definition 3.3. A space is flat if there exists some coordinate system in which the metric becomes

$$g_{mn} = \delta_{mn}.$$

If no coordinate transformation can bring the metric to that form, the space is curved.

That definition preserves exactly the rhythm of the lecture. We may choose complicated coordinates on a flat space, and the metric may then look complicated. But if there is no obstruction to straightening them out and recovering the Kronecker-delta form, then the space is still flat. Curvature is the obstruction to doing that globally.

This mirrors the earlier physics. In the elevator example one can trade apparent gravity for coordinate curvature. But in a real gravitational field there are tidal leftovers. Likewise, in geometry a general metric may look complicated either because the coordinates are complicated or because the space is genuinely curved. The question is whether the complication can be removed.

The lecture then sharpens the local-versus-global distinction. A space may be locally flat without being globally flat. Indeed, in an infinitesimal neighborhood of a point, one can always flatten the metric. But that does not mean that one can do so everywhere at once. One can also have regions of a space that are flat and other regions that are not.

The cone is the most pointed example in the lecture, in both senses. A cone can be sliced and opened out so that any patch away from the tip becomes flat on the page. But the tip remains a special point. It is an obstruction to finding a single global coordinate system in which the whole cone looks Euclidean. The example is useful because it shows vividly that curvature may be concentrated at a place where every sufficiently small patch away from that place looks flat.

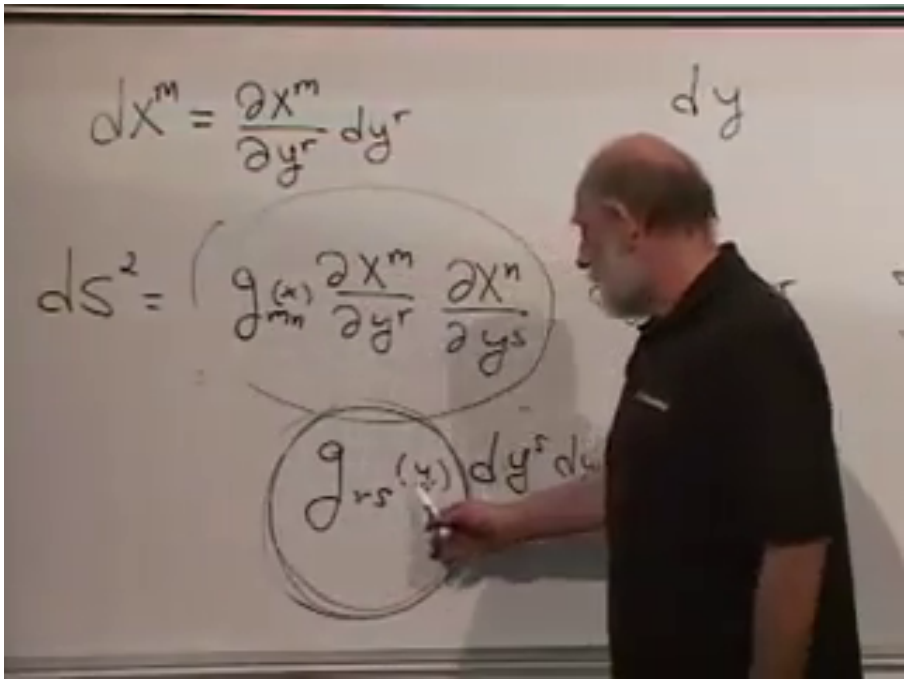


Figure 3.7: Metric tensor under coordinate change. The board circles the coefficient of $dy^r dy^s$ and then identifies it as $g_{rs}^{(y)}$.

3.8 Polar Coordinates as a Worked Flat Example, Then the Sphere

Having discussed the criterion abstractly, Susskind turns to a concrete metric that looks nontrivial but still describes flat space. Take the plane and replace Cartesian coordinates by polar coordinates:

$$x = r\cos\theta, \qquad y = r\sin\theta. \tag{3.20}$$

Differentiate:

$$dx = \cos\theta\, dr - r\sin\theta\, d\theta, \tag{3.21}$$
$$dy = \sin\theta\, dr + r\cos\theta\, d\theta. \tag{3.22}$$

Now substitute these into the flat line element $ds^2 = dx^2 + dy^2$:

$$ds^2 = (\cos\theta\, dr - r\sin\theta\, d\theta)^2 + (\sin\theta\, dr + r\cos\theta\, d\theta)^2 \tag{3.23}$$
$$= \left(\cos^2\theta + \sin^2\theta\right) dr^2 + r^2\left(\sin^2\theta + \cos^2\theta\right) d\theta^2 + 0 \cdot dr\, d\theta \tag{3.24}$$
$$= dr^2 + r^2 d\theta^2. \tag{3.25}$$

So the metric components are

$$g_{rr} = 1, \qquad g_{r\theta} = 0, \qquad g_{\theta\theta} = r^2. \tag{3.26}$$

This is exactly the tension Susskind wants us to feel. The metric is not a constant array of zeros and ones. One component depends on position. Nevertheless the space is flat, because we know the coordinate transformation back to Cartesian coordinates. The metric looks complicated, but the complication can be undone.

3.8.1 Question & Answer

Question. What does the vanishing off-diagonal term actually tell us?

Answer. In this case it tells us something about the coordinates, not by itself about the intrinsic geometry. The lines of constant r are circles, and the lines of constant θ are straight rays from the origin. The absence of a cross term says that these two families of coordinate lines are perpendicular. If we chose a more awkward curvilinear coordinate system on the same flat plane, then a mixed term $g_{r\theta}\, dr\, d\theta$ would generally appear. So $g_{r\theta} = 0$ records an orthogonality property of the chosen coordinates; it is not, by itself, a proof that the space is flat.

This clears the way for the final example. The lecture now turns to the sphere. The transcript becomes badly garbled in the surrounding discussion, so the safest thing is to state only the mathematical content that is solidly indicated. For a unit sphere, in the lecture's late notation, the metric is

$$ds^2 = d\phi^2 + \sin^2\phi\, d\theta^2. \tag{3.27}$$

Here the symbol ϕ is no longer the earlier scalar field; it is now an angular coordinate on the sphere.

The point of the example is straightforward even though the late derivation is not fully recoverable from the transcript: unlike the plane in polar coordinates, the sphere cannot be globally straightened so that its metric becomes the Kronecker delta. No matter which coordinates we choose on the sphere, the obstruction remains. That is curvature.

Remark 3.4. The sphere discussion at the end of the lecture is transcriptually unstable. The metric written above is a cautious standard reconstruction consistent with Susskind's notation and with the surviving mathematical content, but the surrounding derivation should be kept brief.

3.9 Summary

The lecture begins with a physical question about light in an elevator and ends with the metric tensor, and the two halves belong together. A vertical light ray reveals gravitational red shift and blue shift; free fall reveals the exact local cancellation demanded by the equivalence principle; tidal forces reveal the obstruction that remains when one tries to remove gravity globally. That obstruction is then translated into geometry.

From there the mathematical story unfolds step by step. We review the total differential and the chain rule, adopt the summation convention, distinguish scalars from vectors, introduce contravariant and covariant transformation laws, and learn to read the upstairs-downstairs pattern. After the break the metric tensor appears as the coefficient of the line element, and its transformation law shows that it is a covariant tensor of rank 2.

The final payoff is geometrical. A space is flat when some coordinate system restores the metric to the Kronecker-delta form. If no such coordinate system exists globally, there is curvature. Polar coordinates show that a complicated metric need not mean a curved space. The sphere shows the opposite lesson: sometimes the obstruction is real.

Chapter 4

Tensor Algebra, Metrics, and the First Step to Space-Time

This lecture marks a threshold in the course. Susskind says again that he is after the "theoretical minimum," but now adds an important warning: if one were merely to announce the answers in words, that would be less than the minimum. From this point on, the minimum already requires patient calculation. We therefore follow the lecture's progression closely. We begin with transformation laws, not pictures; we build tensor algebra from those laws; we introduce the metric as the first genuinely geometric tensor in the story; and only then do we pivot to flat space, special relativity, and a first time-dependent cosmological metric. These companion notes follow Leonard Susskind's lecture closely and preserve selected documentary board frames associated with LazyingArt LLC and Video2Book.

4.1 Covariant and Contravariant Review

Susskind begins by slowing the room down. We are to review the transformation properties of covariant and contravariant components before trying to do anything more difficult. The reason is not bureaucratic. In curved geometry, and later in space-time, everything depends on knowing exactly what kind of object one is dealing with.

Let x^m and y^m denote two coordinate systems. The convention in the lecture is that coordinates themselves carry upper indices. Once that choice is made, the transformation rules are easy to read.

A covariant object transforms as

$$V_m(y) = \frac{\partial x^n}{\partial y^m} V_n(x), \tag{4.1}$$

while a contravariant object transforms as

$$W^m(y) = \frac{\partial y^m}{\partial x^n} W^n(x). \tag{4.2}$$

This is the point at which Susskind insists that a vector is not "any three numbers" or "anything that can be drawn as an arrow." Temperature, pressure, and humidity give three numbers, but they do not transform as components of a vector. They are just three scalars placed side by side. A vector is defined by its transformation law.

The two basic archetypes come straight from multivariable calculus. If ϕ is a scalar field, then

$$\frac{\partial \phi}{\partial y^m} = \frac{\partial \phi}{\partial x^n} \frac{\partial x^n}{\partial y^m}, \tag{4.3}$$

so the gradient transforms covariantly. And if $y^m = y^m(x)$, then

$$dy^m = \frac{\partial y^m}{\partial x^n}\, dx^n, \tag{4.4}$$

so a differential displacement transforms contravariantly. These are not postulates invented to make tensor calculus elegant. They are chain rules.

It is worth preserving another small but useful convention from the lecture: once the coordinates are written with upper indices, it becomes difficult to get lost. A derivative $\partial x^n/\partial y^m$ naturally carries one upper n and one lower m, and those index positions tell us which kind of object it is prepared to convert into which other kind.

4.1.1 Question & Answer

Question. If an object satisfies one of the vector transformation laws, must it satisfy the other one too?

Answer. Not yet. The lecture is careful about this. At first, a covariant vector and a contravariant vector are simply different kinds of objects. A gradient,

$$\frac{\partial \phi}{\partial x^m},$$

is covariant. A small displacement,

$$dx^m,$$

is contravariant. We are not yet entitled to identify them as two forms of the same vector. That identification only becomes natural once the metric has been introduced and we learn how to raise and lower indices.

4.2 From Vectors to General Tensors

Once we understand the one-index case, the multi-index case is almost forced on us. Each index transforms, and each index brings its own Jacobian factor. For a rank-two covariant tensor,

$$T_{mr}(y) = \frac{\partial x^n}{\partial y^m} \frac{\partial x^s}{\partial y^r} T_{ns}(x). \tag{4.5}$$

For a mixed tensor with one upper and one lower index,

$$T^m{}_n(y) = \frac{\partial y^m}{\partial x^s} \frac{\partial x^r}{\partial y^n} T^s{}_r(x). \tag{4.6}$$

This is the general pattern Susskind wants us to absorb. Indices are not decoration. They are instructions for transformation.

He then turns to the next basic operation, contraction. If one upper and one lower index are identified and summed over, the rank drops. For example,

$$T^m{}_m = T^1{}_1 + T^2{}_2 + T^3{}_3 + \cdots . \tag{4.7}$$

No free index remains, so one expects a scalar. The lecture then checks it explicitly.

If we transform $T^m{}_n$ and then set the two indices equal, the Jacobian factors combine to

$$\frac{\partial x^r}{\partial y^m}\frac{\partial y^m}{\partial x^s} = \delta^r{}_s. \tag{4.8}$$

This is the Kronecker delta, the identity tensor, with one upper and one lower index. It tells us to set $r = s$. Therefore

$$T^m{}_m(y) = T^r{}_r(x), \tag{4.9}$$

and the contracted quantity is indeed a scalar.

The same operation works more generally. If we start with a tensor carrying many upper and lower indices, identify one upper with one lower, and sum, the result is a tensor of lower rank. This is one of the main engines of tensor algebra.

A familiar special case is the inner product. If v^m is contravariant and w_m is covariant, then

$$v^m w_m \tag{4.10}$$

is a scalar.

4.2.1 Question & Answer

Question. If we contract one index of a higher-rank tensor, does that mean the original tensor was simply a scalar times a smaller tensor?

Answer. No. The lecture pauses on exactly this point. Contraction is an operation that produces a new tensor of lower rank. In special cases the new object may factor in a simple way, but in general it does not mean the original tensor was merely a scalar multiple of some simpler one. The contraction throws away structure; it does not reveal a hidden trivial factorization.

This is why contraction matters. It is not just notation. It manufactures coordinate-independent quantities and lower-rank tensors from more complicated ones.

4.3 The Metric Tensor as Distance

At this point the lecture turns from pure index bookkeeping to the first tensor with direct geometric meaning: the metric tensor. If two neighboring points are separated by the infinitesimal displacement dx^m, then the squared distance between them is

$$ds^2 = g_{mn}(x)\, dx^m dx^n. \tag{4.11}$$

The form of this expression is already telling. The two dx's contribute two contravariant indices; the metric contributes two covariant ones; everything is contracted; no free indices remain. Thus ds^2 is a scalar.

Now comes the decisive question: is g_{mn} really a tensor? Susskind does not answer by appeal to intuition. He answers by rewriting the same interval in another coordinate system.

We have

$$dx^m = \frac{\partial x^m}{\partial y^r}\, dy^r. \tag{4.12}$$

Substituting twice into the line element gives

$$ds^2 = g_{mn}(x)\, dx^m dx^n \tag{4.13}$$

$$= g_{mn}(x) \frac{\partial x^m}{\partial y^r} \frac{\partial x^n}{\partial y^s} dy^r dy^s. \tag{4.14}$$

Now define the coefficient of $dy^r dy^s$ to be the metric in the y-coordinates:

$$g_{rs}(y) = \frac{\partial x^m}{\partial y^r} \frac{\partial x^n}{\partial y^s} g_{mn}(x). \tag{4.15}$$

But this is exactly the transformation law for a covariant rank-two tensor. The metric is therefore a tensor in the strict sense used throughout the lecture.

Worked derivation. The derivation is short enough to summarize in the same clipped order in which the lecture presents it:

1. Write the squared distance as $ds^2 = g_{mn}(x)\, dx^m dx^n$.
2. Rewrite each dx^m in terms of dy^r.
3. Substitute and collect the coefficient of $dy^r dy^s$.
4. Call that coefficient $g_{rs}(y)$.
5. Read off the tensor transformation law.

The lecture is careful about what this proves. It proves that the same scalar interval may be expressed in different coordinates using metric components related by the tensor law. It does not mean that two unrelated formulas happen to agree.

4.3.1 Question & Answer

Question. Is ds^2 "the same regardless of the coordinates"?

Answer. Yes, but only in the precise sense just described. We are not computing two different distances and discovering a coincidence. We are taking one and the same scalar interval and rewriting it. That rewriting forces the relation between $g_{mn}(x)$ and $g_{rs}(y)$. The invariance of ds^2 is therefore not a separate afterthought; it is the starting point of the metric transformation law.

4.4 Metric as Matrix, Inverse Metric, and Raising/Lowering

Once the metric has been established as a tensor, the lecture immediately shifts register. A rank-two tensor, viewed at a fixed point and in a fixed coordinate system, is also a matrix. Strictly speaking

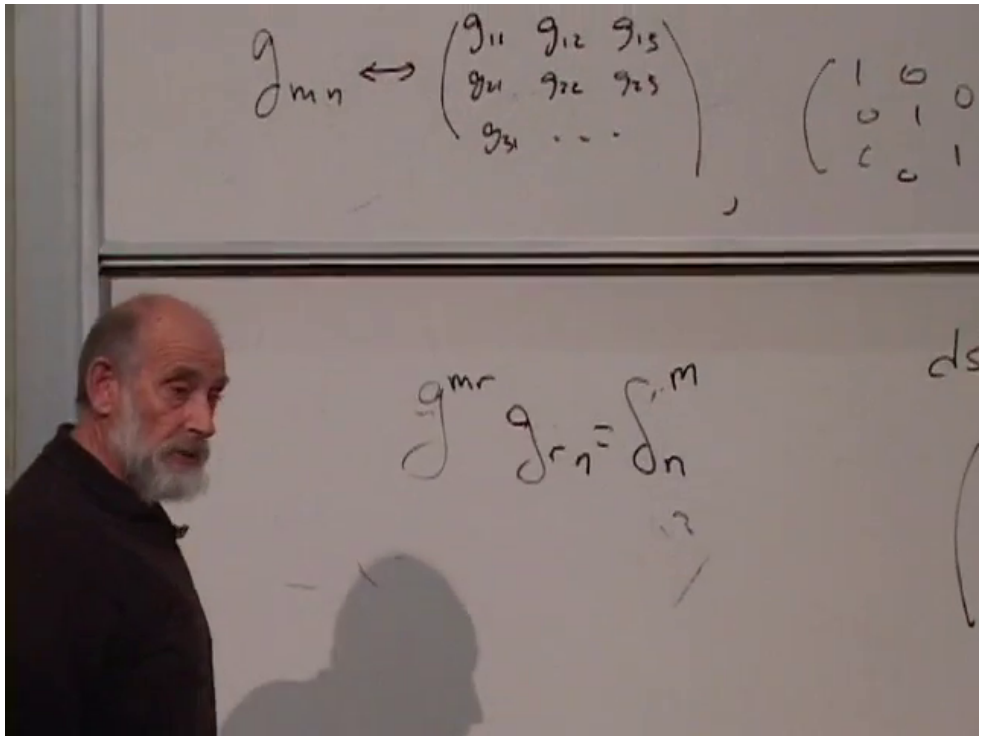


Figure 4.1: Metric tensor as matrix and inverse

the tensor is not the matrix; rather, its components in a given coordinate system form a matrix. But Susskind moves freely between those two descriptions, and the notes should do the same.

Thus we may write

$$g_{mn} \leftrightarrow \begin{pmatrix} g_{11} & g_{12} & g_{13} & \cdots \\ g_{21} & g_{22} & g_{23} & \cdots \\ g_{31} & g_{32} & g_{33} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \tag{4.16}$$

For the line element, the lecture takes the metric to be symmetric,

$$g_{mn} = g_{nm},$$

and then introduces its inverse as an ordinary matrix inverse.

At the matrix level one writes

$$g^{-1} g = I. \tag{4.17}$$

In index notation that becomes

$$g^{mr} g_{rn} = \delta^{m}{}_{n}. \tag{4.18}$$

Here the Kronecker delta must carry one upper and one lower index. The lecture explicitly stops to correct the temptation to write both indices downstairs. In tensor notation $\delta^{m}{}_{n}$ is the identity tensor.

The upper-index object g^{mn} is then interpreted as the collection of components of the inverse matrix. This is not a new geometric structure carrying new information; it is another representation

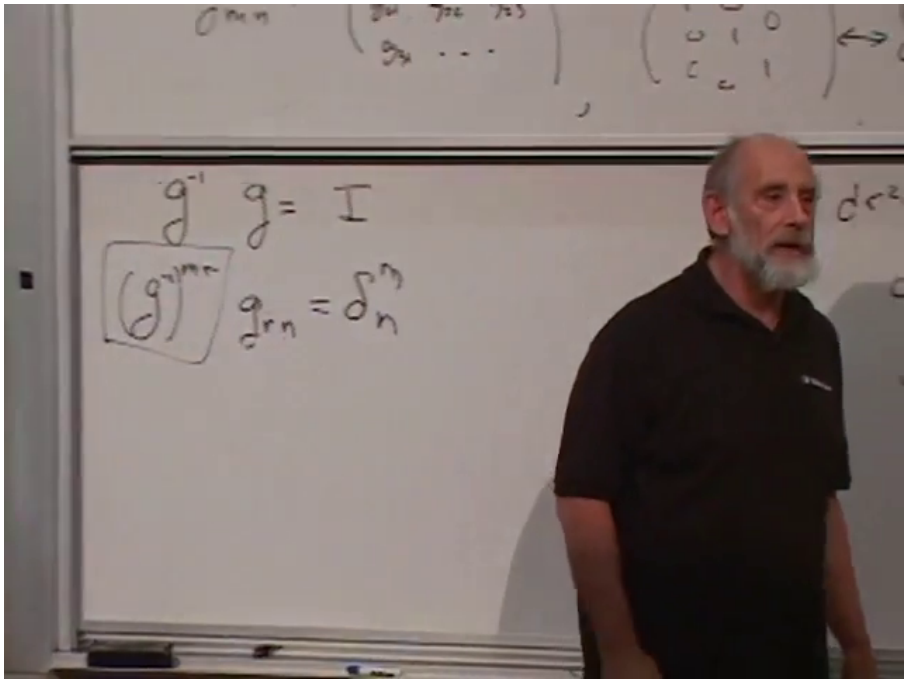


Figure 4.2: Inverse metric contraction identity

of the same metric. The nontrivial theorem, stated but not proved in the lecture, is that these inverse-matrix components themselves transform tensorially, now as a contravariant rank-two tensor.

It is therefore legitimate to speak of the metric with lower indices and the metric with upper indices. They contain the same information.

The lecture also gives a simple worked example. In polar coordinates on the plane,

$$ds^2 = dr^2 + r^2 d\theta^2. \tag{4.19}$$

If r is the first coordinate and θ the second, then

$$g_{11} = 1, \qquad g_{22} = r^2, \tag{4.20}$$

with no off-diagonal terms. The inverse metric is therefore

$$g^{11} = 1, \qquad g^{22} = \frac{1}{r^2}. \tag{4.21}$$

For a diagonal matrix, the inverse is found by inverting the diagonal entries.

4.4.1 Question & Answer

Question. Can the metric tensor fail to have an inverse?

Answer. A general tensor can fail to have an inverse, but the lecture insists that the metric is not of that kind. Susskind attributes this to symmetry together with the absence of zero eigenvalues, and then postpones the proof. For the purposes of the lecture, invertibility is part of the working definition of the metric structure we are using.

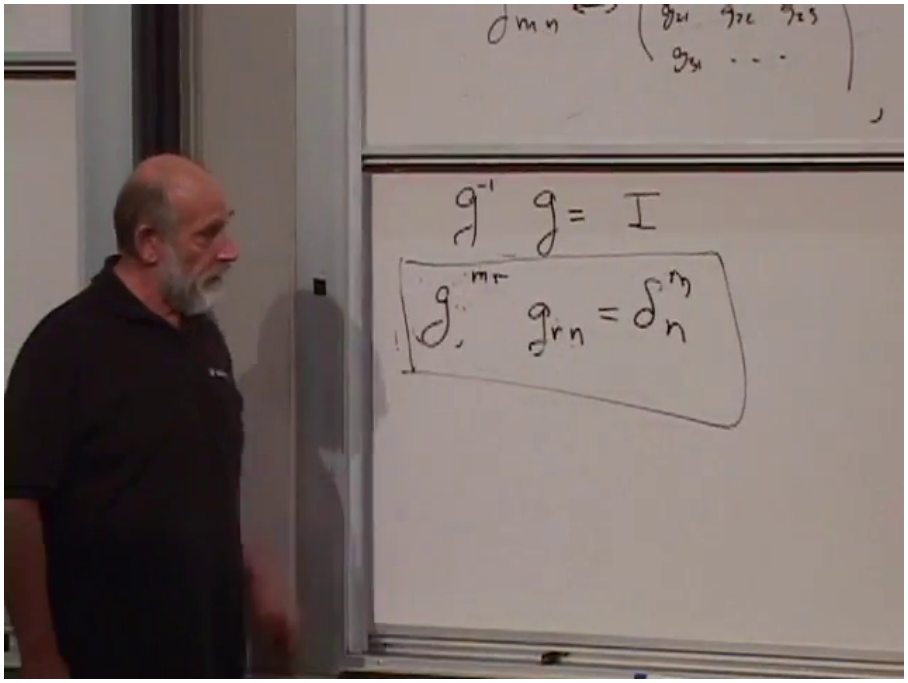


Figure 4.3: Inverse metric relation

4.4.2 Question & Answer

Question. Why may we take the metric to be symmetric?

Answer. Because only the symmetric part contributes to the quadratic form ds^2. The cross terms appear as

$$g_{12}\, dx^1 dx^2 + g_{21}\, dx^2 dx^1 = (g_{12} + g_{21})\, dx^1 dx^2, \tag{4.22}$$

since $dx^1 dx^2 = dx^2 dx^1$. The antisymmetric part cancels out of the line element. There is therefore no loss of generality, for the purposes of distance, in taking the metric symmetric.

Now comes the first serious use of the metric. Given contravariant components V^m, we define the corresponding covariant components by

$$V_n = g_{nm} V^m. \tag{4.23}$$

Conversely,

$$V^m = g^{mn} V_n. \tag{4.24}$$

Likewise, for a rank-two tensor,

$$T^m{}_n = g^{mr} T_{rn}. \tag{4.25}$$

This is the operation of raising and lowering indices.

The lecture then applies the same idea to the differential displacement:

$$dx_m = g_{mn}\, dx^n. \tag{4.26}$$

The board at this stage is pedagogically valuable because the circled factor marks the lowering step. It is also slightly misleading: the first chalk line suggests a literal equality that Susskind later

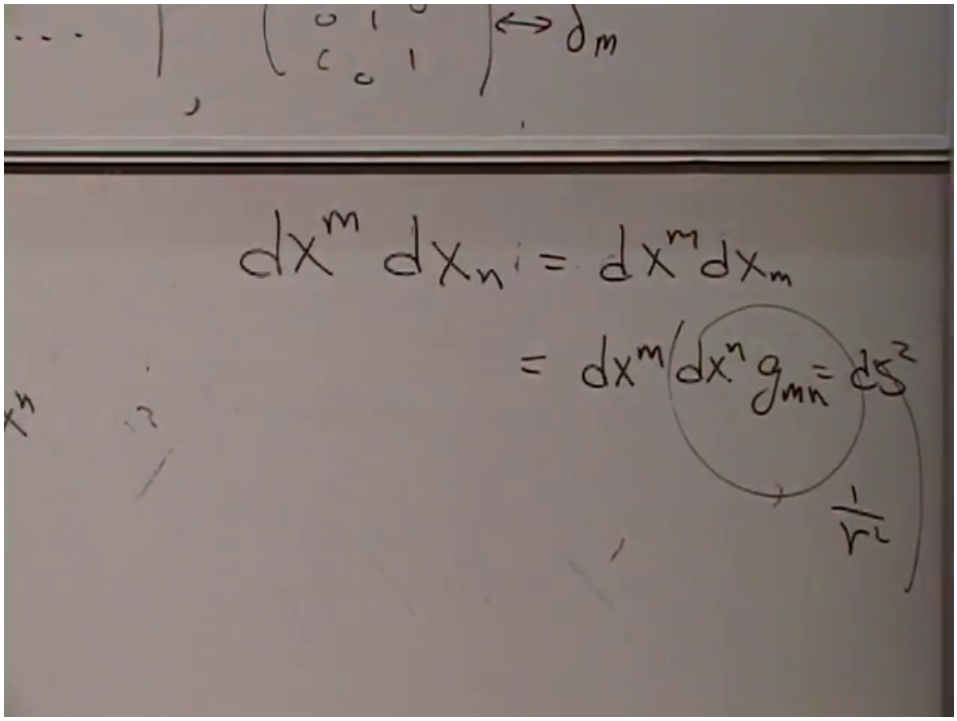


Figure 4.4: Lowering an index with the metric

withdraws. What is intended is a contraction step, not an equality between unlike expressions. Written cleanly, the argument is

$$dx^m dx_m = dx^m (g_{mn} dx^n) \tag{4.27}$$
$$= g_{mn}\, dx^m dx^n \tag{4.28}$$
$$= ds^2. \tag{4.29}$$

There is also a third equivalent form:

$$g^{mn} dx_m dx_n = ds^2. \tag{4.30}$$

So the same scalar interval may be written in three ways:

$$ds^2 = g_{mn}\, dx^m dx^n = dx^m dx_m = g^{mn} dx_m dx_n. \tag{4.31}$$

This is the moment at which the early separation between covariant and contravariant components softens. Once the metric is given, the lecture is perfectly willing to speak of upper and lower components of the same vector.

4.5 Tensor Equalities and the End of Tensor Algebra

Having developed the metric and its inverse, the lecture briefly returns to the general algebra of tensors and extracts two useful principles.

The first is a warning. Tensors may be added only if they are of the same type. It makes sense to add T_{mn} and W_{mn}. It does not make sense to add, say, a tensor with lower indices to one with

upper indices and expect the result to transform coherently. The index structure is part of the identity of the tensor.

The second principle is much more powerful. If a tensor is zero in one coordinate system, it is zero in every coordinate system. The reason is immediate from the transformation law: the transformed components are Jacobian factors multiplying the original components. If the original components are zero, the transformed components are zero.

From this it follows that if two tensors are equal in one coordinate system, they are equal in every coordinate system. Indeed, if

$$T = W$$

in one coordinate system, then

$$T - W = 0$$

there; but a zero tensor is zero everywhere, so $T = W$ everywhere.

This is why tensor equations are so powerful. Each side may have complicated coordinate dependence in its separate components, but if the two sides are equal once, they remain equal after every legitimate coordinate transformation. The equality is geometric, not coordinate-bound.

At this point the lecture gestures toward tensor calculus and then refuses to begin it. Ordinary derivatives of tensor components are not, in general, tensor components. One needs a more careful notion of differentiation. Susskind postpones that subject to the next lecture, and the postponement is deliberate. He wants the class to leave tensor algebra with a clean pivot into space-time.

4.6 Flat Space, Cartesian Frames, and the Pivot to Space-Time

The re-entry into geometry is through the simplest case: ordinary flat space. Even if we restrict ourselves entirely to Cartesian coordinates, there is still more than one Cartesian coordinate system. We may rotate the axes; we may translate them; and even if the origin is held fixed, there is still freedom to orient the coordinate frame differently.

In two-dimensional Cartesian coordinates the line element is

$$ds^2 = (dx^1)^2 + (dx^2)^2 = \delta_{mn}\, dx^m dx^n. \tag{4.32}$$

Within this restricted class of Cartesian-to-Cartesian transformations there is no distinction between upper and lower indices, because the metric is just the identity. The lecture does, however, insert a caution: writing δ_{mn} with both indices downstairs is tolerable only in this restricted flat-Cartesian setting. It is not the correct fully tensorial notation in general coordinates.

The board picture here is especially useful because it puts formula and geometry side by side. On the left is the Euclidean interval; on the right is a second coordinate description of the same plane. The mathematical statement is that the allowable Cartesian transformations are precisely those that preserve the form of the Euclidean interval. In that sense they preserve the form of Pythagoras' theorem.

A schematic redraw helps make the coordinate geometry legible while keeping the original board evidence nearby.

Only after making these remarks about ordinary space does Susskind pivot to special relativity. A naive Euclidean guess in one time and one space dimension would be

$$dt^2 + dx^2.$$

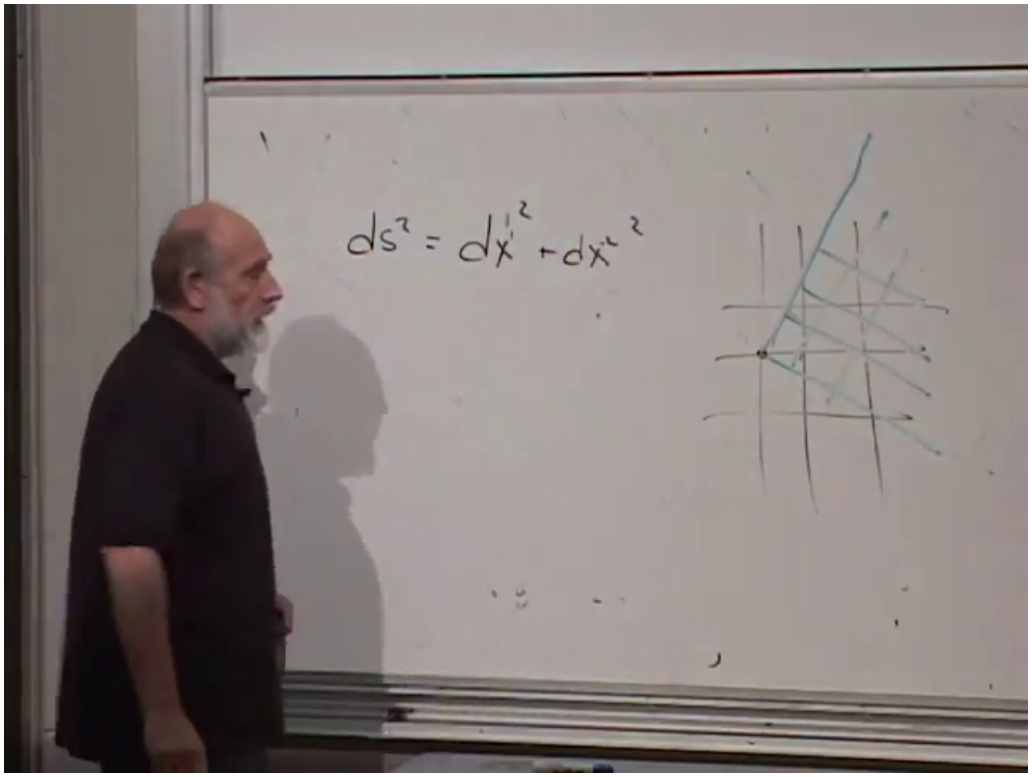


Figure 4.5: Flat line element and skew grid

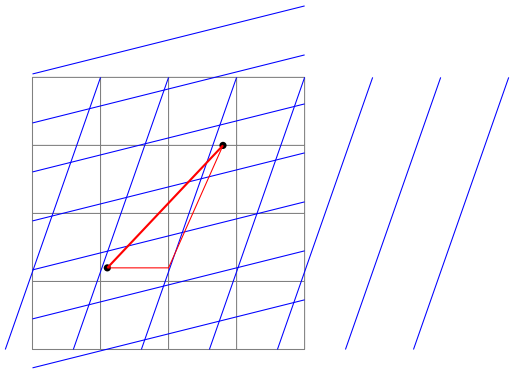


Figure 4.6: Schematic redraw of the flat-plane picture: an orthogonal grid together with a skew coordinate net describing the same plane, and a highlighted interval whose Euclidean length is preserved.

But this is not the quantity preserved by Lorentz transformations. The invariant interval is

$$d\tau^2 = dt^2 - dx^2. \tag{4.33}$$

In three spatial dimensions,

$$d\tau^2 = dt^2 - dx^2 - dy^2 - dz^2. \tag{4.34}$$

The lecture then pauses over units. Setting $c = 1$ is not just a convenience. It is the analog of measuring all spatial directions in the same units. If we insisted on measuring space and time in mismatched units, the formulas would be littered with conversion factors. In relativity, c is that conversion factor, and choosing $c = 1$ removes an artificial asymmetry.

Special relativity is then summarized in one line: it is the theory of coordinate transformations that preserve the form of this indefinite interval.

4.7 General Space-Time Metric and the First Cosmological Example

Now the lecture is ready to move from space to space-time. We switch from Latin to Greek indices and write the four coordinates as

$$x^\mu, \qquad \mu = 0, 1, 2, 3,$$

with x^0 the time coordinate. This is the lecture's convention, and it is worth keeping because the notation itself helps mark the transition from spatial tensors to space-time tensors.

The natural generalization of the special-relativistic interval is exactly what the earlier tensor algebra suggests:

$$d\tau^2 = g_{\mu\nu}(x)\, dx^\mu dx^\nu. \tag{4.35}$$

At this stage one need not yet invoke genuine gravitation. One may already obtain nontrivial coordinate dependence by introducing curvilinear coordinates into flat space-time. The formalism comes first; the full gravitational interpretation comes later.

The flat special case is Minkowski space. In Minkowski coordinates the metric is

$$\eta_{\mu\nu} = \mathrm{diag}(1, -1, -1, -1), \tag{4.36}$$

so that

$$d\tau^2 = \eta_{\mu\nu}\, dx^\mu dx^\nu. \tag{4.37}$$

The symbol $\eta_{\mu\nu}$ is the space-time analog of δ_{mn}, except that its signature is not all plus. In the convention of this lecture there is one positive eigenvalue and three negative ones. Because the diagonal entries are ± 1, the inverse is the same matrix:

$$\eta^{\mu\nu} = \eta_{\mu\nu}. \tag{4.38}$$

The tensor rules themselves are unchanged. Contravariant objects still transform with $\partial y/\partial x$, covariant ones with $\partial x/\partial y$. The only novelty is the signature of the metric and the geometric meaning of the resulting scalar interval.

4.7.1 Question & Answer

Question. What does it mean, in the language of the metric, to say that space-time is expanding?

Answer. It means that the metric coefficients depend on time in such a way that the spatial geometry changes with time. The lecture writes the diagonal matrix rather loosely, with entries $1, -A(t), -A(t), -A(t)$, but describes the interval as having a squared scale factor. To match the interval as stated, the clean normalization is

$$d\tau^2 = dt^2 - a^2(t)\big(dx^2 + dy^2 + dz^2\big), \tag{4.39}$$

where $a(t)$ is the scale factor. If $a(t)$ grows, then at fixed coordinate separations the spatial part of the metric grows, and that is what the lecture means here by an expanding universe.

Susskind then builds intuition by analogy rather than by curvature calculations. In two-dimensional space we already know

$$ds^2 = dr^2 + r^2 d\theta^2, \tag{4.40}$$

which is just the plane written in polar coordinates. If we replace r^2 by $\sin^2 r$, we obtain

$$ds^2 = dr^2 + \sin^2 r\, d\theta^2, \tag{4.41}$$

the metric of a sphere. If instead we take

$$ds^2 = dr^2 + e^{2r} d\theta^2, \tag{4.42}$$

we obtain a horn-like surface whose circumference grows exponentially with r.

The lecture's point is not yet to compute curvature. It is to teach us how to read geometry directly from metric coefficients. In the surface examples the geometry changes as a function of r; in the cosmological example the geometry changes as a function of t. That is the first concrete meaning of "expanding space-time" in the language of the metric tensor.

4.8 Summary

This lecture does two things at once. First, it completes the tensor algebra that general relativity cannot do without: transformation laws, mixed tensors, contraction, the metric, the inverse metric, and the raising and lowering of indices. Second, it uses exactly that machinery to pivot from ordinary space to space-time. Flat Euclidean geometry becomes Minkowski geometry, and Minkowski geometry becomes the general space-time line element

$$d\tau^2 = g_{\mu\nu}(x)\, dx^\mu dx^\nu.$$

The lecture ends by postponing tensor calculus and curvature, but that postponement is itself part of the structure. We leave this chapter with the algebraic machinery in hand and with the first metric picture of an expanding universe already on the board.

Chapter 5

Tensor Calculus, Connections, and Geodesics

Susskind begins by warning us that the subject is slippery, abstract, and a little difficult. That warning is useful. The lecture is not merely a list of formulas. It is a careful shift from tensor algebra to tensor calculus, from knowing how tensors transform to knowing how to differentiate them without losing the coordinate-independence that made them useful in the first place. We first remind ourselves why tensors matter, then we watch the ordinary derivative fail for vectors, and only then do we introduce the connection and the covariant derivative. After that the lecture resets with a concrete polar-coordinate example, uses the metric to identify the Christoffel symbols, and finally turns toward curves, geodesics, and the gravitational interpretation.

5.1 Why Tensor Calculus Enters General Relativity

We have already learned most of what tensor algebra can do for us. Compatible tensors can be added and subtracted: covariant components with covariant components, contravariant components with contravariant components, but not mismatched objects. That reminder is brief, and it serves only to clear the ground. The real question is why we introduce tensors at all.

The answer is geometric and practical at the same time. Geometric facts do not depend on coordinates. The Earth is almost spherical, the surface has bumps and curvature, Mount Everest is a lump on that surface: none of these statements requires coordinates to be true. But when we actually compute, we do use coordinates. We use components. Computation is about numbers, and tensors are the way geometry is translated into numbers without surrendering the geometry.

So the goal is this: write equations whose numerical components may change from one coordinate system to another, but whose truth does not. Susskind makes the point in two equivalent ways. If two tensorial quantities are equal in one coordinate system, they are equal in all coordinate systems. And if a tensor is zero in one coordinate system, it is zero in all coordinate systems. This is why it is so natural to write laws in the form "something equals zero."

To remind ourselves, consider a covariant vector:

$$V_n(y) = \frac{\partial x^r}{\partial y^n} V_r(x). \tag{5.1}$$

If all components $V_r(x)$ vanish, then all components $V_n(y)$ vanish. The same is true for a mixed tensor:

$$T^m{}_n(y) = \frac{\partial y^m}{\partial x^r}\frac{\partial x^s}{\partial y^n}T^r{}_s(x). \tag{5.2}$$

Again, zero in one frame means zero in every frame.

This is the basic promise of tensors. General relativity will need that promise not just for algebraic relations, but for differential equations. The difficulty is that ordinary derivatives do not automatically preserve it.

5.2 Scalars First, Then the Trouble with Vectors

The lecture wisely begins with the easiest case. A scalar field ϕ does not really transform at all. It is simply the same number attached to the same point, whatever coordinates we use to label that point. Its derivative therefore behaves well under a change of coordinates:

$$\frac{\partial \phi}{\partial y^n} = \frac{\partial x^m}{\partial y^n}\frac{\partial \phi}{\partial x^m}. \tag{5.3}$$

So the derivative of a scalar transforms as a covariant vector.

This is also exactly what intuition suggests. If the temperature in a room is constant everywhere, then it is constant in Cartesian coordinates, in polar coordinates, and in any other coordinates one cares to invent. If $\partial\phi/\partial x^m = 0$ everywhere, then $\partial\phi/\partial y^n = 0$ everywhere as well. For scalars, the derivative behaves exactly as we hoped.

Now we turn to vectors, and the lecture deliberately creates some tension. Imagine a vector field in ordinary flat space that points in the same direction everywhere and has the same length everywhere. In Cartesian coordinates its components are constant from point to point, so their derivatives vanish. So far, so good.

But now lay a curvilinear coordinate grid on top of the same flat blackboard. The physical vector field is unchanged. What changes is the coordinate frame. The axes bend; they rotate from point to point. A component is a projection onto those local axes, so it depends not only on the vector itself but also on the orientation of the axes. A geometrically constant vector field can therefore have nonconstant components in curvilinear coordinates.

This is the first real obstruction. The derivative of a scalar detects genuine variation of the field. The derivative of vector components can also detect the motion of the coordinates themselves. That extra sensitivity is exactly what must be repaired.

5.3 Testing the Ordinary Derivative of a Vector

Susskind now slows the lecture down and performs the central calculation. We assume, temporarily, that the ordinary derivative of a vector component might transform as a tensor, and then we watch that assumption fail.

Define

$$T_{mn}(x) \equiv \frac{\partial V_m(x)}{\partial x^n}. \tag{5.4}$$

Suppose this really were the x-components of a tensor. Then in the y-coordinates it would have to transform as

$$T_{mn}(y) = \frac{\partial x^r}{\partial y^m}\frac{\partial x^s}{\partial y^n}T_{rs}(x) \tag{5.5}$$

$$= \frac{\partial x^r}{\partial y^m}\frac{\partial x^s}{\partial y^n}\frac{\partial V_r(x)}{\partial x^s}. \tag{5.6}$$

Now the last two factors are exactly arranged for the chain rule, so this becomes

$$T_{mn}(y) = \frac{\partial x^r}{\partial y^m}\frac{\partial V_r(x)}{\partial y^n}. \tag{5.7}$$

At this stage the lecture pauses and asks whether this is the same thing as the ordinary derivative of the transformed vector components. So let us calculate that directly. Since

$$V_m(y) = \frac{\partial x^r}{\partial y^m}V_r(x), \tag{5.8}$$

we get

$$\frac{\partial V_m(y)}{\partial y^n} = \frac{\partial}{\partial y^n}\left(\frac{\partial x^r}{\partial y^m}V_r(x)\right) \tag{5.9}$$

$$= \frac{\partial x^r}{\partial y^m}\frac{\partial V_r(x)}{\partial y^n} + \frac{\partial}{\partial y^n}\left(\frac{\partial x^r}{\partial y^m}\right)V_r(x). \tag{5.10}$$

The first term is exactly what we wanted. The trouble is the second term:

$$\frac{\partial}{\partial y^n}\left(\frac{\partial x^r}{\partial y^m}\right)V_r(x). \tag{5.11}$$

Thus

$$\frac{\partial V_m(y)}{\partial y^n} = T_{mn}(y) + \frac{\partial}{\partial y^n}\left(\frac{\partial x^r}{\partial y^m}\right)V_r(x), \tag{5.12}$$

and therefore

$$\frac{\partial V_m(y)}{\partial y^n} \neq T_{mn}(y) \tag{5.13}$$

in general.

This is the core calculation of the lecture. The ordinary derivative of vector components fails the tensor test because the derivative does not only act on the field. It also acts on the coordinate-transformation factor.

5.3.1 Question & Answer

Why did the scalar case work, while the vector case failed?

Because a scalar carries no changing basis information with it. A vector does. When we rewrite $V_m(y)$ in terms of $V_r(x)$, we must also rewrite how the basis itself is sitting inside the geometry. The factor $\partial x^r/\partial y^m$ records that basis change. When the derivative hits that factor, it measures the fact that the coordinates are flopping around from point to point.

This also explains why the trouble disappears if the coordinate transformation is linear. In that case $\partial x^r/\partial y^m$ is constant, its derivative vanishes, and ordinary differentiation behaves as if the bases were rigid. So the failure is not mysterious. It is exactly the mathematical trace of a moving coordinate frame.

5.4 Covariant Derivatives of Vectors and Tensors

Once the failure has been exposed, the next move becomes natural. The ordinary derivative misses tensoriality by a definite extra term, so we try to repair the derivative by adding a correction of the same general form. That correction is called Γ.

This is the point at which the lecture stabilizes its new object on the board:

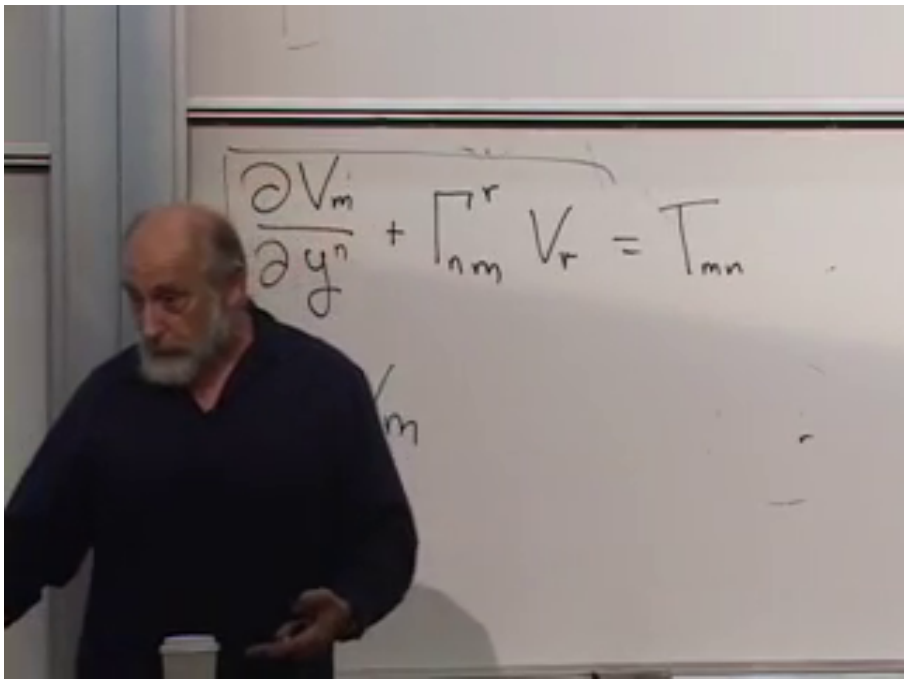


Figure 5.1: Covariant derivative of a lower-index vector

The cleaned version of the board equation is

$$\nabla_n V_m = \frac{\partial V_m}{\partial y^n} + \Gamma^r{}_{nm} V_r. \tag{5.14}$$

The same line is also read as saying that the corrected derivative is a rank-two tensor:

$$\frac{\partial V_m}{\partial y^n} + \Gamma^r{}_{nm} V_r = T_{mn}. \tag{5.15}$$

Remark 5.1. The board formula for the lower-index vector carries a plus sign in front of the connection term, and that sign is clearly visible in the extracted frames. We retain that local lecture convention when quoting this stage of the argument.

Susskind immediately emphasizes that "covariant derivative" does not mean "derivative of a covariant vector." It is an unfortunate overlap of words. The phrase names the operation, not the index type of the thing to which it is applied.

Now the lecture broadens the rule. If we differentiate a tensor with two lower indices, each lower index contributes its own correction term. The right way to think about it is exactly the way Susskind explains it in class: turn a blind eye to all but one index, pretend the visible index belongs

to a vector, write the corresponding correction, and then repeat for the other indices. Thus for a rank-two covariant tensor,

$$\nabla_p T_{mn} = \frac{\partial T_{mn}}{\partial y^p} + \Gamma^r{}_{pm} T_{rn} + \Gamma^r{}_{pn} T_{mr}. \tag{5.16}$$

The lecture then revisits the one-index example after relabeling the derivative index to p:

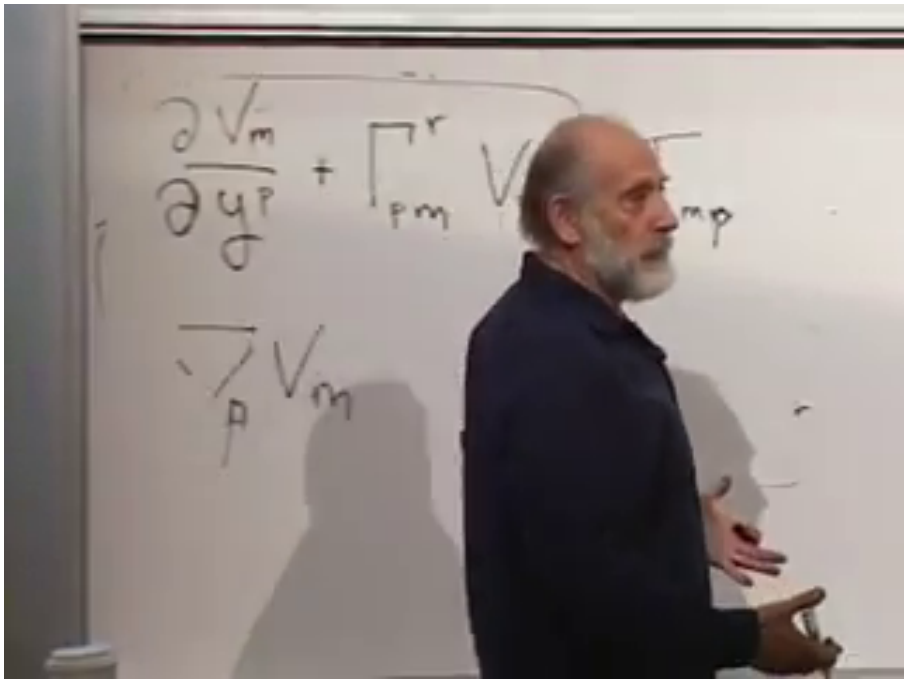


Figure 5.2: Reindexed covariant-derivative formula

$$\nabla_p V_m = \frac{\partial V_m}{\partial y^p} + \Gamma^r{}_{pm} V_r. \tag{5.17}$$

The right-hand side tensor label is partly occluded in the frame, but the intended reading is that this object is the tensor T_{mp}.

The lecture later also writes the corresponding formula for an upper-index vector:

$$\nabla_m V^n = \frac{\partial V^n}{\partial y^m} + \Gamma^n{}_{mr} V^r. \tag{5.18}$$

This will be the form we need when we come to derivatives along curves.

5.4.1 Question & Answer

At this point the audience raises a very good bookkeeping question: may we interchange a free index with a repeated one?

No. A repeated index is a dummy summation index, and dummy indices may be renamed. A free index is not a dummy index, and in general it may not be swapped with another free index or with

a repeated one. One may only interchange indices when there is an independent symmetry of the tensor that justifies it.

So the rule is:

- repeated indices may be renamed;
- free indices may not be interchanged merely for convenience;
- if a tensor is symmetric or antisymmetric in certain slots, then one may use that symmetry, but not otherwise.

That warning matters because the entire middle part of the lecture is about index bookkeeping. The formulas are only trustworthy if the indices are treated honestly.

5.5 A Concrete Example: Polar Coordinates

After the abstract discussion, Susskind resets with a concrete example. Go to polar coordinates and ignore the origin, which he explicitly calls a nasty point. Consider a vector field with

$$V_\theta = 0, \qquad V_r = 1. \tag{5.19}$$

Its polar components are constant: one and zero everywhere. But the vector field itself is not constant, because at every point it points radially outward from the origin. The arrow on the right side of the plane is different from the arrow at the top, and different again from the arrow on the left.

So here we have exactly the kind of example the lecture wanted. The components are constant, yet the vector field is clearly varying from place to place. In Cartesian coordinates those components would not be constant. In polar coordinates they are constant. The geometric field is the same in both descriptions. Therefore it is the covariant derivative, not the ordinary derivative of components, that captures the real variation.

In standard polar coordinates on the flat plane,

$$ds^2 = dr^2 + r^2\, d\theta^2. \tag{5.20}$$

Thus the metric has a nonconstant component, namely $g_{\theta\theta} = r^2$. That is enough to generate nontrivial Christoffel symbols even though the underlying space is flat. So flatness does not mean that the Christoffel symbols vanish in every coordinate system. It means that one can find coordinates, namely Cartesian coordinates, in which they vanish.

5.6 Metric Compatibility and the Christoffel Symbols

Up to this point Γ has been introduced formally, as the thing that must be added to the ordinary derivative in order to make a tensor. The lecture now asks what determines Γ geometrically.

The answer comes from the metric. In flat Cartesian coordinates,

$$g_{mn} = \delta_{mn}, \tag{5.21}$$

so the metric is constant from point to point. If the covariant derivative is really the correct derivative, then the covariant derivative of the metric ought to vanish there. But the covariant derivative of the metric is supposed to be a tensor. Therefore, if it vanishes in one coordinate system, it must vanish in all coordinate systems. This motivates metric compatibility:

$$\nabla_r g_{mn} = 0. \tag{5.22}$$

With the lecture's local sign convention for lower indices, the expansion reads

$$\frac{\partial g_{mn}}{\partial y^r} + \Gamma^s{}_{rm} g_{sn} + \Gamma^s{}_{rn} g_{ms} = 0. \tag{5.23}$$

This is now an equation for the connection in terms of the metric and its derivatives.

The board derivation of the solution is noisy in the transcript, and Susskind himself tells the audience that the algebra is tedious and that they should check the final answer at home. So here it is best to state the cleaned endpoint in its familiar normalized form:

$$\Gamma^A{}_{BC} = \frac{1}{2} g^{AD} \left(\partial_B g_{DC} + \partial_C g_{DB} - \partial_D g_{BC} \right). \tag{5.24}$$

This is the Christoffel symbol. Susskind also calls it the connection, or affine connection. The important point in the lecture is not the memorization of the index pattern, but the fact that the connection is built from the metric and its first derivatives.

It is equally important that the Christoffel symbols are not tensors. In flat space with Cartesian coordinates, all derivatives of the metric vanish and therefore $\Gamma^A{}_{BC} = 0$. But if we change to curvilinear coordinates in the same flat space, derivatives of the metric need not vanish, so the Christoffel symbols need not vanish. A genuine tensor that is zero in one frame would be zero in all frames. Hence the Christoffel symbols are not tensors.

The lecture then adds a geometric interpretation that is worth preserving. In any Riemannian space, curved or not, if we look at a small enough patch we can introduce coordinates as close to Cartesian as possible. At a chosen point the metric can be made equal to δ_{mn}, and its first derivatives can also be made to vanish there. At that point the Christoffel symbols vanish, and the covariant derivative reduces to the ordinary derivative. So the covariant derivative tells us how a tensor varies in coordinates that are, at the point in question, as flattened out as possible.

5.7 Differentiation Along a Curve

Now the lecture changes scale. Up to here we have differentiated tensor fields from point to point in space. The next problem is to differentiate along a particular curve.

Take a curve and parameterize it by arc length s. The first step is to understand the scalar case. If ϕ is a scalar field, then its derivative along the curve is

$$\frac{d\phi}{ds} = \frac{\partial \phi}{\partial y^m} \frac{dy^m}{ds}. \tag{5.25}$$

The factor

$$t^m \equiv \frac{dy^m}{ds} \tag{5.26}$$

is the tangent vector to the curve. Because s is arc length, t^m is a unit vector tangent to the curve.

This is a nice moment in the lecture. Susskind pauses to make the geometry visible. The vector dy^m points along the curve, and dividing by the infinitesimal length ds produces the unit tangent vector. So the scalar derivative along the path is just the gradient contracted with the tangent direction.

For a vector field the same plan must be corrected. The ordinary derivative of the components along the curve is not tensorial for the same reason as before: the coordinates may be moving around. So we simply replace the ordinary derivative by the covariant derivative. The covariant derivative of a vector along the curve is defined by

$$\frac{DV^n}{Ds} = \nabla_m V^n \frac{dy^m}{ds}. \tag{5.27}$$

Expanding $\nabla_m V^n$, we find

$$\frac{DV^n}{Ds} = \frac{dV^n}{ds} + \Gamma^n{}_{mr} V^r \frac{dy^m}{ds}. \tag{5.28}$$

The first term is the ordinary derivative along the curve. The second term accounts for the fact that the coordinate frame itself may be turning as we move.

This is exactly the pattern the lecture wanted us to see: take the ordinary derivative, then add the Γ-term, and finally contract with the tangent vector to specify the direction of differentiation.

5.8 Geodesics and the Road Toward Gravity

Now we go one step further and differentiate the tangent vector itself. Put

$$V^n = t^n = \frac{dy^n}{ds}. \tag{5.29}$$

Then the covariant derivative of the tangent vector along the curve is

$$\frac{Dt^n}{Ds} = \frac{d^2y^n}{ds^2} + \Gamma^n{}_{mr} \frac{dy^m}{ds}\frac{dy^r}{ds}. \tag{5.30}$$

In flat space with Cartesian coordinates, a straight line has constant tangent vector, so this quantity is zero. But the lecture now asks the more interesting question: can this still happen in a genuinely curved space? Can there be curves in a curved space whose tangent vector is covariantly constant?

The answer is yes.

Definition 5.2. A geodesic is a curve whose tangent vector is covariantly constant. Equivalently,

$$\frac{Dt^n}{Ds} = 0, \tag{5.31}$$

or in second-order form,

$$\frac{d^2y^n}{ds^2} + \Gamma^n{}_{mr} \frac{dy^m}{ds}\frac{dy^r}{ds} = 0. \tag{5.32}$$

This is not yet curvature theory, but it is already the geometry of motion. A geodesic is the straightest possible curve in the space. In coordinates that are locally as flat as possible, its tangent vector has constant components for a short distance, until the coordinates begin to bend again.

The lecture then explicitly bridges to physics. In special relativity, straight lines in space-time are the world-lines of particles on which no force acts. In general relativity, free particles move on geodesics. The geodesic equation therefore plays the role of an equation of motion. The left-hand side looks like an acceleration term, and the connection term looks like a force term. Since the Christoffel symbols are built from derivatives of the metric, the derivatives of the metric become the relativistic analog of the gravitational force.

Susskind is careful here: he is not yet proving the full gravitational dynamics. He is telling us where the formalism is going. We already have the apparatus needed to talk about motion. What comes next is parallel transport, and then curvature, the obstruction to flattening a space globally.

5.9 Summary

This lecture unfolds a single argument step by step. We begin with the reason tensors matter: they let us write equations whose truth survives a change of coordinates. The derivative of a scalar respects that principle, but the ordinary derivative of a vector does not, because the derivative sees the variation of the coordinates themselves. The product rule exposes the failure explicitly.

From that failure the cure emerges. We introduce the connection, build the covariant derivative, and then generalize it index by index. The polar-coordinate example shows why constant components are not enough. Metric compatibility then ties the connection to the metric and leads to the Christoffel symbols, which are not tensors but are exactly the objects needed to differentiate tensors correctly.

Finally, by contracting the covariant derivative with a tangent vector, we learn how to differentiate along a curve. Setting the covariant derivative of the tangent vector to zero gives the geodesic equation. At that point the geometry is already leaning toward physics: the Christoffel symbols, hence derivatives of the metric, enter as the force-like terms governing the motion of free particles in a gravitational field.

Chapter 6

Proper Time, Local Flatness, and Curvature from Parallel Transport

The lecture begins by announcing two long-range goals: to say more about geodesics, and to connect geodesics more directly with gravity. But it does not move there immediately. First the space-time metric has to be rebuilt, because without proper time, signature, and covariant differentiation already in hand, the later discussion of curvature would float free of its physical meaning. Only after that reset does the lecture choose, after a brief hesitation, to take up curvature first. The result is a chapter whose logic is quite deliberate: proper time, then the distinction between curved coordinates and genuine curvature, then local flatness, then geodesics and parallel transport, and only then curvature as the failure of parallel transport to be path-independent.

6.1 Space-Time Metric, Proper Time, and the Minkowski Signature

For the first part of the lecture we are asked to think about space-time rather than space alone. The mathematical machinery is the same as before, but the notation shifts. Instead of the spatial indices $m, n, \ldots$, Susskind now uses Greek indices $\mu, \nu, \ldots$, running over $0, 1, 2, 3$, with 0 the time direction and $1, 2, 3$ the spatial directions.

There is, again, a metric, and therefore a notion of interval. In space-time the interval is written as proper time:

$$d\tau^2 = g_{\mu\nu}\, dx^\mu dx^\nu, \tag{6.1}$$

$$\eta_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \tag{6.2}$$

Here $\eta_{\mu\nu}$ is the special-relativistic metric, constant everywhere, and the lecture uses the signature $(+---)$: one plus sign for the time direction and three minus signs for the spatial ones.

Strictly speaking, one should not say that the metric simply *is* this matrix once and for all. The metric is a tensor. What is true is that in these coordinates its components take this form. That

is exactly why the lecture pauses over the point that the metric tensor is a tensor, even in this simplest case.

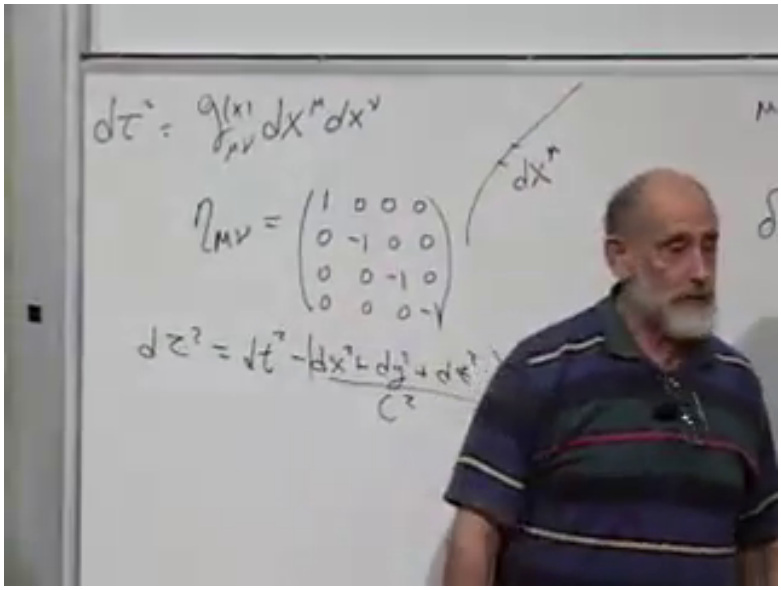


Figure 6.1: Minkowski metric and the proper-time interval, written on the board from top to bottom as general contraction, explicit matrix, and component form.

The last line on the board is slightly cramped, but the intended formula is clear from the lecture:

$$d\tau^2 = dt^2 - \frac{dx^2 + dy^2 + dz^2}{c^2}. \tag{6.3}$$

If we choose units with $c = 1$, this reduces to

$$d\tau^2 = dt^2 - dx^2 - dy^2 - dz^2. \tag{6.4}$$

The infinitesimal displacement dx^μ is the little step along the worldline, and $d\tau$ is the corresponding interval measured along that worldline.

The lecture also pauses over the phrase that the speed of light is "large." Of course c is neither large nor small in itself. What is meant is that the speeds of the objects we care about are small compared with c. Under those circumstances the spatial correction is negligible, and so

$$d\tau \approx dt \tag{6.5}$$

for slow motion. But that is only an approximation. In general the proper time along the trajectory is not the same thing as the coordinate time of a chosen frame.

6.1.1 Question & Answer

Why does the moving clock record $d\tau$ rather than dt? Because the clock moves with the object. It does not measure the time coordinate of some externally chosen system of coordinates; it measures the interval along the object's own worldline. The lecture makes the point with a geometric analogy: a tape measure laid along a curve in space does not record x or y, it records the distance *along the curve*. In the same spirit, the moving clock is a kind of space-time tape measure. It marks off proper time along the trajectory itself. Coordinate time dt belongs to a coordinate system; proper time $d\tau$ belongs to the motion.

6.2 Curved Coordinates Are Not Yet Curvature

With the interval and proper time back in place, the lecture makes the bridge to general relativity by analogy. Flat space is to curved space as special relativity is to curved space-time. The formal tools are the same: metric, tensors, Christoffel symbols, covariant derivatives. What changes is that the metric need no longer be constant, and therefore the geometry itself may be nontrivial.

But there is a distinction the lecture is determined not to lose. Curved coordinates are not yet the same thing as genuine curvature.

In flat space or flat space-time one does not ordinarily need curvilinear coordinates at all. If our only goal were to make life easy, introducing them would seem perverse. We introduce them because accelerated frames are physically important. In an inertial frame objects move on straight lines; in an accelerated frame the same motions look curved, and the coordinate lines themselves are bent.

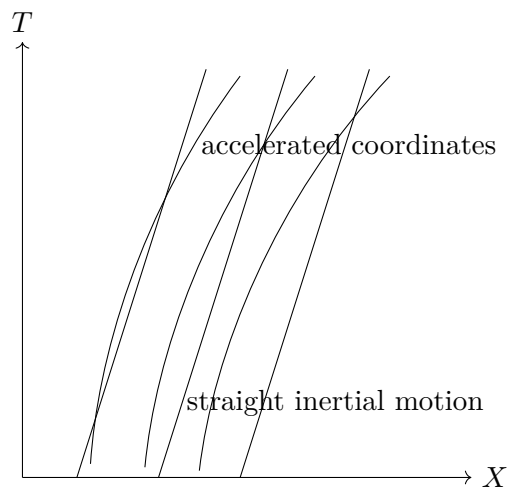


Figure 6.2: A schematic Minkowski diagram: straight inertial trajectories in (T, X), together with a curved coordinate family adapted to accelerated motion.

This is the old elevator story. An accelerated coordinate system can imitate a gravitational field. That is the content of the equivalence principle in its local form. Had Einstein not taken acceleration and gravity seriously together, nobody would have volunteered curved coordinates in otherwise flat physics.

But this imitation is only local. A fake gravitational field produced by acceleration can be removed by changing coordinates again. A real gravitational field cannot be removed everywhere in that way.

The lecture's example is a star pulling an elevator toward it. If the elevator is small and allowed to fall freely, experiments inside it look inertial. That is the local victory of free fall. But the victory is not complete. The field varies slightly across the elevator, and so tidal forces remain. One side of the elevator wants to fall differently from another. That residual variation is the first sign that something genuine is present. It is the obstruction to removing the gravitational field globally.

At this point, after a brief hesitation and even a vote in the room, the lecture decides to do curvature before returning to geodesics. That is the right order for the story. Before we can say how things move in curved space-time, we must first say what it means for the geometry itself to be flat or curved.

6.3 What Does It Mean for Space or Space-Time to Be Flat?

Now the discussion shifts back to ordinary geometric notation, with Latin indices $m, n, r, s, \ldots$. The first question is the obvious one: what does it mean for a space, or a space-time, to be flat?

The answer is not that every coordinate system looks Cartesian. The answer is that one *can find* coordinates in which the metric is constant. In the simplest Euclidean case one may choose coordinates with

$$g_{mn} = \delta_{mn},$$

but the lecture immediately points out that even this is stronger than necessary. If the coordinate axes are straight and uniformly spaced but not mutually perpendicular, the metric will not be δ_{mn}, yet it will still be constant from point to point. So the real definition is constancy, not orthonormality.

To put it slightly differently, a geometry is flat if one can choose coordinates so that the derivatives of the metric vanish:

$$\partial_s g_{mn} = 0. \tag{6.6}$$

This is already a three-index object, with m and n symmetric. In a truly flat geometry one can make all such derivatives vanish everywhere.

The lecture then turns the statement local. Even without having defined curvature formally, one can already assert that at a *single point* it is always possible to choose coordinates so that the first derivatives of the metric vanish there. That is the mathematical counterpart of the freely falling elevator. One can always flatten the geometry locally. What curvature will measure is the obstruction to doing this everywhere at once.

There is also an implicit regularity assumption here. The lecture says it plainly when a student asks about wildly wiggly metrics. We assume the metric is continuous and differentiable, at least in the region under discussion. There can be exceptions, and those exceptions are genuine singularities. But one must not confuse a singular coordinate system with a singular geometry. Polar coordinates are bad at the origin, yet the plane itself is perfectly smooth there. A true singularity is a place where no coordinate system makes the metric smooth. The cone tip will supply the example.

6.4 Local Coordinate Freedom and Vanishing First Derivatives

The local flattening argument is one of the lecture's technical centers. We begin with coordinates x^m in which the metric is not locally constant at the chosen point. We now look for new coordinates y^m with the same origin and write a local expansion

$$y^m = x^m + c^m{}_{rs}\, x^r x^s + \cdots . \tag{6.7}$$

The linear term has been chosen to be the identity simply to keep the discussion focused. The real freedom lies in the quadratic coefficients $c^m{}_{rs}$. These are the knobs we adjust in order to influence the first derivatives of the transformed metric.

The condition we want is

$$\frac{\partial g_{mn}(y)}{\partial y^s} = 0 \tag{6.8}$$

at the chosen point. Why should this be possible? The lecture answers not by performing the full transformation, but by counting equations and unknowns.

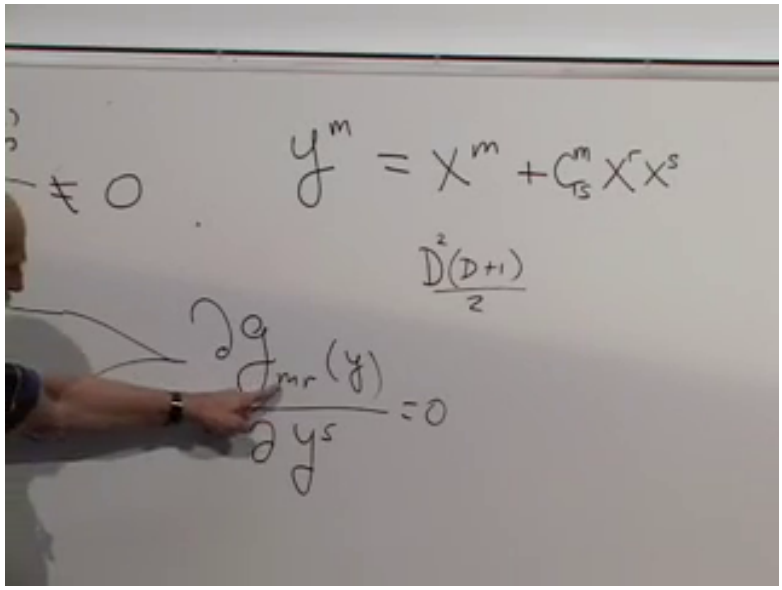


Figure 6.3: The board layout makes the argument visual: the quadratic coordinate freedom sits above, and the target condition on the first derivatives of the metric sits below.

First, the coefficients ${c^m}_{rs}$ may be taken symmetric in r and s:

$$ {c^m}_{rs} = {c^m}_{sr}, \tag{6.9} $$

since $x^r x^s = x^s x^r$. For fixed m, that means ${c^m}_{rs}$ is a symmetric $D \times D$ matrix, and therefore has

$$ \frac{D(D+1)}{2} \tag{6.10} $$

independent entries. In three dimensions that gives 6, which is exactly the arithmetic checked aloud in the lecture.

Worked derivation.

1. For fixed m, the coefficients ${c^m}_{rs}$ form a symmetric matrix in the indices r, s.
2. A symmetric $D \times D$ matrix has $D(D+1)/2$ independent entries.
3. Since m itself runs over D values, the total number of independent quadratic coefficients is

$$ D \cdot \frac{D(D+1)}{2} = \frac{D^2(D+1)}{2}. \tag{6.11} $$

4. The object $\partial_s g_{mn}$ also has three indices, with m, n symmetric.
5. Therefore the number of first-derivative conditions matches the number of adjustable quadratic coefficients.
6. So there is just enough local coordinate freedom to choose coordinates in which all first derivatives of the metric vanish at the selected point.

This is deliberately a local statement. The lecture is explicit that the same counting does not continue indefinitely. Once one asks to eliminate second derivatives, third derivatives, and so on, there are simply too many independent conditions. Thus one can always kill the first derivatives at a point, but in general not the higher derivatives. That is precisely why the higher-order structure contains real geometry.

6.4.1 Question & Answer

Why is there enough coordinate freedom to kill the first derivatives, but not all higher derivatives? Because the quadratic part of the coordinate transformation carries exactly the same index structure as the first derivative of the metric: a three-index object with two symmetric slots. That makes the counting come out exactly right at first order. But higher derivatives bring in more independent components than the available coordinate freedom can absorb. So the lecture's conclusion is not that a curved geometry can be made flat, but only that every geometry can be made *locally* flat through first order at a point.

6.5 Christoffel Symbols and Metric Compatibility

Once the first derivatives of the metric have been set to zero at a point, the next step follows at once. The Christoffel symbols are built from those first derivatives:

$$\Gamma^r{}_{mn} = \frac{1}{2} g^{rs} \left(\frac{\partial g_{sm}}{\partial x^n} + \frac{\partial g_{sn}}{\partial x^m} - \frac{\partial g_{mn}}{\partial x^s} \right). \tag{6.12}$$

So if

$$\frac{\partial g_{mn}(x)}{\partial x^r} = 0 \tag{6.13}$$

at the point under discussion, then immediately

$$\Gamma^r{}_{mn} = 0 \tag{6.14}$$

there as well.

This observation already carries one important lesson. Christoffel symbols cannot be tensors. A true tensor that is nonzero somewhere cannot be made to vanish at a point merely by changing coordinates, but the Christoffel symbols can. The lecture says this explicitly: the ability to transform them away at a point is itself evidence that they are not tensorial objects.

The next move is the familiar one:

$$\nabla_r g_{mn} = 0. \tag{6.15}$$

In components, the covariant derivative of the metric takes the form

$$\nabla_r g_{mn} = \partial_r g_{mn} - \Gamma^s{}_{rm} g_{sn} - \Gamma^s{}_{rn} g_{ms}. \tag{6.16}$$

At the chosen point, in the locally flattened coordinates, the ordinary derivative term vanishes and the Christoffel terms vanish as well. Hence $\nabla g = 0$ there. But ∇g is a tensor. If a tensor vanishes in one coordinate system at a point, it vanishes in every coordinate system at that point. Since the point was arbitrary, the standard conclusion is that the metric is covariantly constant everywhere.

This is also the place where the lecture reiterates the original purpose of the covariant derivative. The ordinary derivative of a tensor is not a tensor. The Christoffel symbols are not tensors. But the right combination of the two produces an object that really is tensorial.

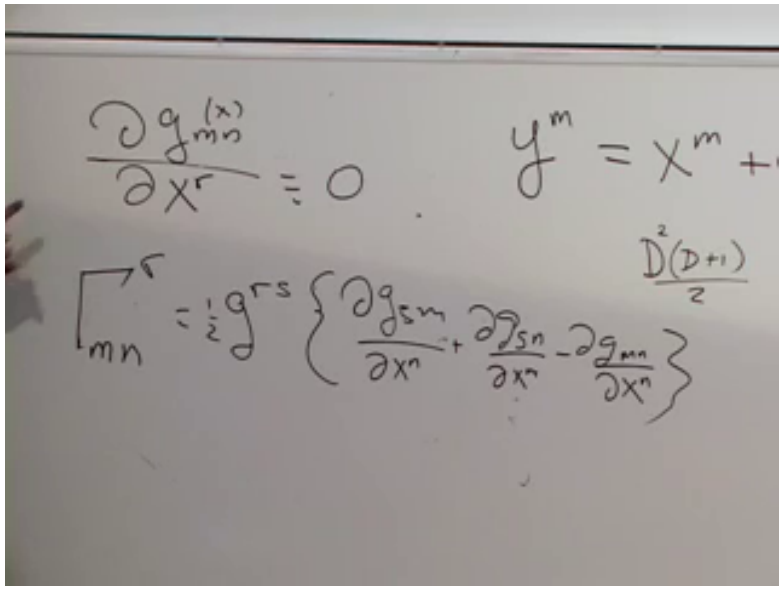


Figure 6.4: The board now turns the counting argument into a geometric statement: vanishing first derivatives imply vanishing Christoffel symbols at the chosen point.

6.5.1 Question & Answer

Is the argument for $\nabla g = 0$ circular? A student raises exactly that objection, and Susskind grants the point in part. The argument is a little circular. What it really establishes is the consistency of the standard construction. One assumes the usual Christoffel symbols built from the metric, defines covariant differentiation in the usual way, and then checks that with those definitions the metric is covariantly constant. That is not presented as an immaculate theorem from nowhere. It is presented as the standard, internally consistent choice, and the notes should keep that modest register.

6.6 Covariant Derivative Along a Curve, Geodesics, and Parallel Transport

Only after this detour through local coordinates does the lecture return to the first topic it announced at the beginning: curves. The starting point is the covariant derivative of a vector field v^m along a curve $x^m(s)$. Here s is simply a parameter along the curve; in the lecture it is thought of as distance along the curve, but the essential formulas do not depend on the special choice of parameter.

The tangent vector is

$$t^p = \frac{dx^p}{ds}, \tag{6.17}$$

and the covariant derivative of v^m along the curve is

$$\frac{Dv^m}{Ds} = \frac{dv^m}{ds} + \Gamma^m{}_{np}\, v^n \frac{dx^p}{ds}. \tag{6.18}$$

The first term by itself is not tensorial, because the coordinate axes can vary from point to point. The Christoffel term corrects for that change.

A geodesic is now recovered as a special case. We set the vector equal to the tangent vector itself,

$$v^m = \frac{dx^m}{ds},$$

and demand that this tangent vector be parallel transported along the curve:

$$\frac{D}{Ds}\left(\frac{dx^m}{ds}\right) = 0. \tag{6.19}$$

Then

$$\frac{d^2x^m}{ds^2} + \Gamma^m{}_{np}\frac{dx^n}{ds}\frac{dx^p}{ds} = 0. \tag{6.20}$$

This is the geodesic equation in the form used throughout the lecture. A geodesic is the straightest possible curve in the geometry. In flat space it is an ordinary straight line; on a sphere it is a great circle; in a general curved geometry it is whatever path results from going incrementally straight ahead.

But the lecture does not linger there. It immediately widens the discussion. Parallel transport applies to any vector, not only to tangent vectors and not only along geodesics. If a vector v^m along a chosen curve satisfies

$$\frac{Dv^m}{Ds} = 0, \tag{6.21}$$

then we say that v^m is parallel transported along that curve. In words, this is as close as one can come, in a curved geometry or a curved coordinate system, to saying that the vector is constant.

The lecture then rewrites the condition in incremental form. Multiplying by ds, we obtain

$$dv^m = -\Gamma^m{}_{np}\, v^n\, dx^p. \tag{6.22}$$

This is the useful update rule. Once the vector is known at one point, together with the Christoffel symbols and the infinitesimal displacement dx^p, it tells us how to advance the vector to the neighboring point so that the covariant derivative continues to vanish.

Worked derivation.

1. Start with the parallel-transport condition

$$\frac{Dv^m}{Ds} = 0.$$

2. Expand the covariant derivative:

$$\frac{dv^m}{ds} + \Gamma^m{}_{np}\, v^n \frac{dx^p}{ds} = 0.$$

3. Multiply by ds.
4. Rearranging gives

$$dv^m = -\Gamma^m{}_{np}\, v^n\, dx^p.$$

5. The parameter s has disappeared; what remains is the infinitesimal update along the actual curve through the differentials dx^p.

The lecture is very concrete here. A gyroscope provides the model. Carry it by its tip along a path, do not forcibly twist it, and it continues to point in the same direction. That is a physical implementation of parallel transport. It is also why the lecture says that parallel transport preserves the angles between vectors and, in fact, their magnitudes as well, although the direction is the point of interest here.

6.6.1 Question & Answer

Does parallel transport mean keeping a fixed angle to the curve? No. The lecture goes out of its way to correct this misunderstanding. If we transport a vector around an ordinary circle in flat space while keeping it parallel to itself, its angle relative to the tangent of the circle changes continuously. At one point it may be perpendicular to the tangent; later it may line up with it. What remains fixed is not the angle to the curve, but the infinitesimal relation between the vector and its neighboring transported value.

There is one important special case. In two dimensions, if the curve itself is a geodesic, then a vector parallel transported along it does keep a fixed angle with the tangent. The lecture mentions that fact, and then deliberately refuses to turn it into a general discussion of torsion or twisting in higher dimensions. Here the point is simpler: geodesics are one special instance of parallel transport, not the whole story.

6.7 Curvature as Path Dependence of Parallel Transport

Now the lecture asks its central question. Suppose we start with a vector at one point and parallel transport it to another point. If we use one curve, we get some final answer. If we use a different curve, must we get the same answer? The sharpest version of the question is to use a closed curve. If we transport the vector all the way around a loop, does it come back to itself?

In flat space, yes. In a curved space, not in general. That failure is the operational definition of curvature. A geometry is curved if there are closed curves around which a parallel transported vector does not return to its original direction.

Before giving the model example, the lecture inserts an important distinction. A bad coordinate system is not the same thing as a bad geometry. Polar coordinates are singular at the origin, but the plane is not. A genuine singularity is a point where no coordinate system makes the metric smooth. The cone tip is the example to keep in mind.

The cone is produced by cutting an angular wedge out of a flat sheet and identifying the cut edges. Away from the tip the geometry is flat. Only the tip is special. The lecture uses this in exactly the right way: first to show how curvature may be concentrated at a point, then to show what parallel transport actually detects.

If the loop stays away from the tip, the relevant region can be opened out onto the flat sheet, and the transported vector comes back to itself. But if the loop encloses the tip, the identification of the cut edges forces a mismatch. The vector comes back rotated by the angular deficit:

$$\Delta\theta = \theta. \tag{6.23}$$

The striking feature is that the detailed shape of the loop does not matter. What matters is whether the loop winds around the tip. The same point of curvature rotates the vector by the same amount.

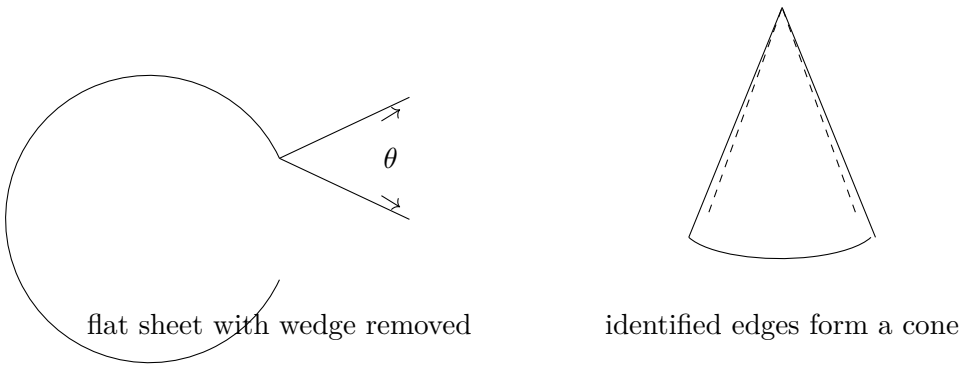


Figure 6.5: The cone as a flat sheet with an angular deficit and the corresponding identified surface.

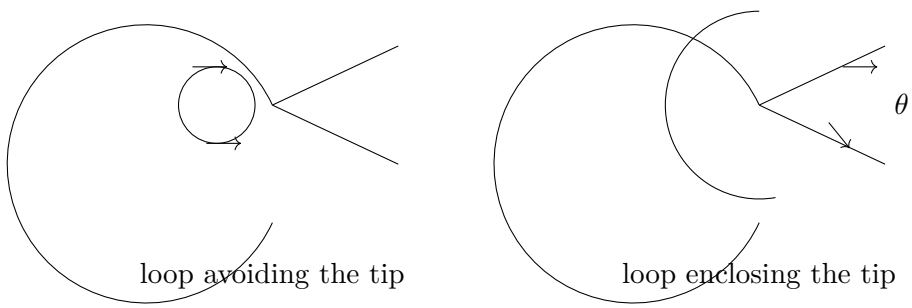


Figure 6.6: A loop that avoids the cone tip returns the vector unchanged; a loop that encloses the tip returns it rotated by the angular deficit.

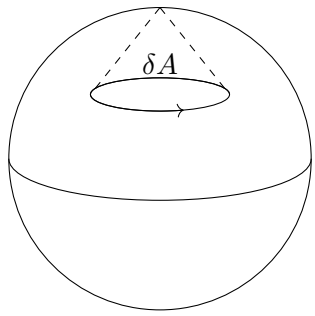
The lecture then sharpens the example. Suppose we smooth off the cone tip instead of leaving it singular. A large loop around the same region will still detect curvature inside. So the closed-loop test locates the presence of curvature in the interior, but not necessarily its detailed distribution. On the sharp cone the curvature is concentrated at a single point. On the smoothed cone it is spread over a small cap. The loop knows there is curvature in there; it does not yet know precisely how it is distributed.

There is another benefit to the cone. It lets us think about geodesics by cutting and pasting. In the opened flat picture, geodesics are just straight lines. If a line crosses the cut, we simply re-open the cone elsewhere so that the same intrinsic geodesic is represented by a straight line on the new flat chart. This is the lecture's preferred way to think about geodesics on the cone: not by forcing them into the original slit representation, but by moving the slit until the straightness becomes visible.

The sphere supplies the smooth version of the same idea. Take a small loop on a sphere. The lecture asks us to imagine a cone tangent to the sphere all along that loop. A tiny creature living on the surface and carrying a little gyroscope cannot distinguish, by purely local measurements along the loop, whether it lives on the sphere or on that tangent cone. Therefore the transported vector undergoes the same net rotation in both cases.

For small loops the rotation is small, and the lecture states the local two-dimensional law

$$\delta\theta \propto \delta A. \tag{6.24}$$



A small loop on the sphere and the tangent-cone picture behind the area law.

Figure 6.7: For a sufficiently small loop on a smooth surface, the transport defect is controlled by the enclosed area.

More precisely, one defines the local curvature R by

$$\delta\theta = R\,\delta A \tag{6.25}$$

for an infinitesimal loop. This is not yet the general curvature tensor of four-dimensional space-time. It is the local curvature law for a two-dimensional surface. It says that curvature measures the infinitesimal rate at which parallel transport around a tiny loop rotates a vector.

The lecture is careful about the domain of this formula. It is a small-loop statement. For large loops one needs the full integral effect, not merely the infinitesimal coefficient. That is why the formula should be read as local.

6.7.1 Question & Answer

Do the transported vectors have to lie in the surface, and is parallel transport just keeping them parallel in the embedding space? No on both counts, at least not in the naive sense. On a two-dimensional surface the vector is best thought of as an element of the tangent space at each point. It points along the surface, but it is not literally a little stick glued into some ambient three-dimensional space. That is why parallel transport is an intrinsic notion. It is not defined by keeping the vector parallel in the larger embedding space and then projecting back. If one did that, the whole intrinsic notion of curvature would be lost. The lecture returns to this point near the end: intrinsic parallel transport lives in the tangent spaces of the geometry itself.

The lecture next introduces the sign of curvature. Give the enclosed area a sign by orienting the boundary of the loop. If we traverse the loop counterclockwise, say, we call the area positive; the opposite orientation is negative. Positive curvature means that the transported vector rotates in the same sense as the oriented loop, while negative curvature means it rotates in the opposite sense:

$$\operatorname{sign}(\delta\theta) = \operatorname{sign}(\delta A) \qquad \text{for positive curvature.} \tag{6.26}$$

The sphere is the model of positive curvature. A saddle surface is the model of negative curvature. The lecture even gives a cut-and-paste version of negative curvature: instead of removing a wedge, insert an extra wedge. Then a counterclockwise circuit makes the vector rotate clockwise. The inner region of an ordinary torus is of this saddle type, while the outer region is sphere-like and positively curved. The torus is therefore useful as a mixed example: positive curvature outside,

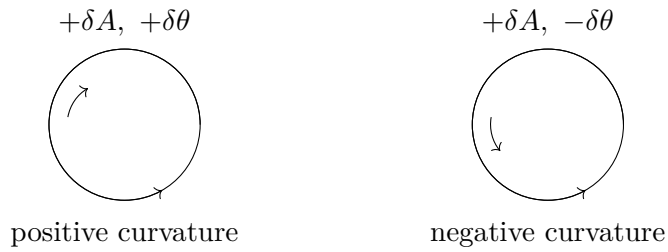


Figure 6.8: For an oriented loop, positive curvature rotates the transported vector in the same sense as the loop, while negative curvature rotates it in the opposite sense.

negative curvature inside. The Mobius strip is mentioned as a global oddity, but the lecture clearly treats it as a side remark rather than part of the main mathematical spine.

Finally the discussion lifts back to space-time. The words are the same, even if the visualization is not. There are curves in space-time, there is parallel transport, and the failure of a vector to return to itself around a closed loop is still the sign of curvature. The real gravitational field is precisely this space-time curvature, not merely the effect of funny coordinates. Mass, and more generally energy and momentum, create curvature. Particles then move on geodesics of the curved space-time. In the appropriate limit those geodesics reproduce Newtonian motion, with small relativistic corrections.

That is where the lecture stops. The next step, deferred to the following lecture, is to convert this geometric picture into the curvature tensor itself.

6.8 Summary

The lecture does not begin with curvature tensors or field equations. It begins more patiently: with the space-time interval, proper time, and the Minkowski signature. From there it separates curved coordinates from genuine curvature by way of the accelerated elevator and tidal forces. Flatness is then defined as the ability to choose coordinates in which the metric is constant, or equivalently has vanishing first derivatives. At a single point this can always be done, and the Christoffel symbols can be made to vanish there as well. That local flattening is the mathematical form of the free-fall principle.

Only after that does the lecture return to curves. The tangent vector gives the geodesic equation, and the more general condition $Dv^m/Ds = 0$ defines parallel transport for any vector along any curve. The central test for curvature is then path dependence: transport a vector around a closed loop and see whether it comes back unchanged. The cone gives the sharp example, the sphere the smooth one, and the small-loop law $\delta\theta = R\,\delta A$ gives the local two-dimensional measure.

The final physical point is the one general relativity needs. Space-time curvature is the genuine gravitational field. Energy and momentum create that curvature, and particles move on the resulting geodesics. The next lecture will give that idea its tensorial form.